

Association of Hemispheric Laterality with Neurosurgery in ICH: A Post-Hoc Bayesian Analysis of ATACH-2 and ERICH

Details of Statistical Analysis

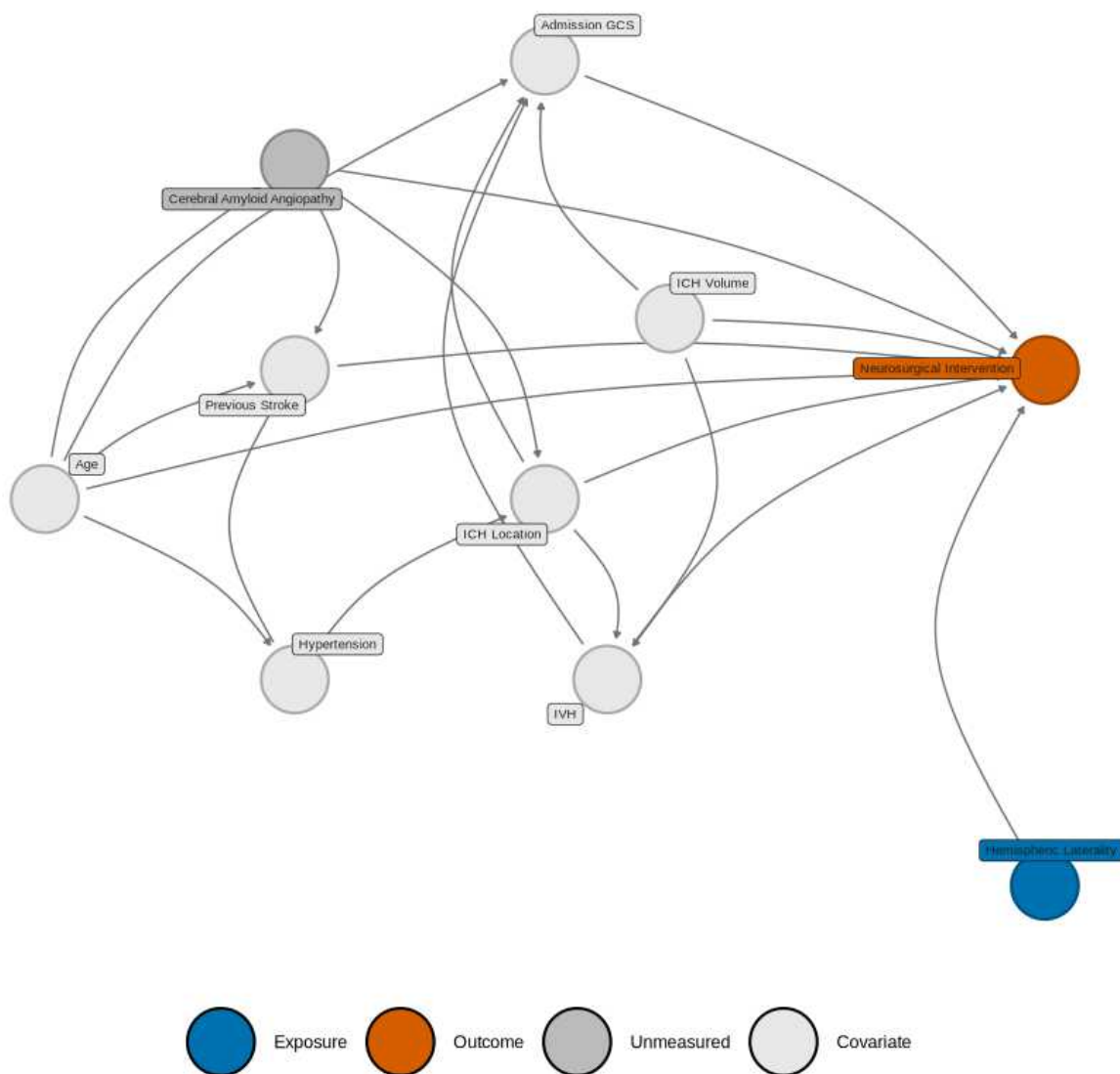
Covariate Selection

Primary and Secondary Outcomes

For our primary outcome model we specified the following DAG:

Neurosurgical Intervention

Directed acyclic graph for the primary surgical outcome



For neurosurgical intervention and all acute care secondary outcomes, the target estimand was the total effect of hemispheric laterality — the overall causal effect of laterality on the outcome, encompassing all pathways through which it operates. To reduce bias from measured confounding, we adjusted for baseline variables that independently influence both the clinical presentation associated with hemispheric laterality and the likelihood of intervention: age,

admission Glasgow Coma Scale (GCS) score, ICH location, baseline ICH volume, and presence of intraventricular hemorrhage (IVH). These covariates act as confounders in the causal structure and were included to block backdoor paths, not to decompose direct and indirect effects.

We chose not to adjust for National Institutes of Health Stroke Scale (NIHSS) for 2 reasons: 1) it is strongly correlated with GCS as both represent quantitative representations of the neurological exam, and 2) NIHSS is partly determined by lateralized deficits such as aphasia and neglect.

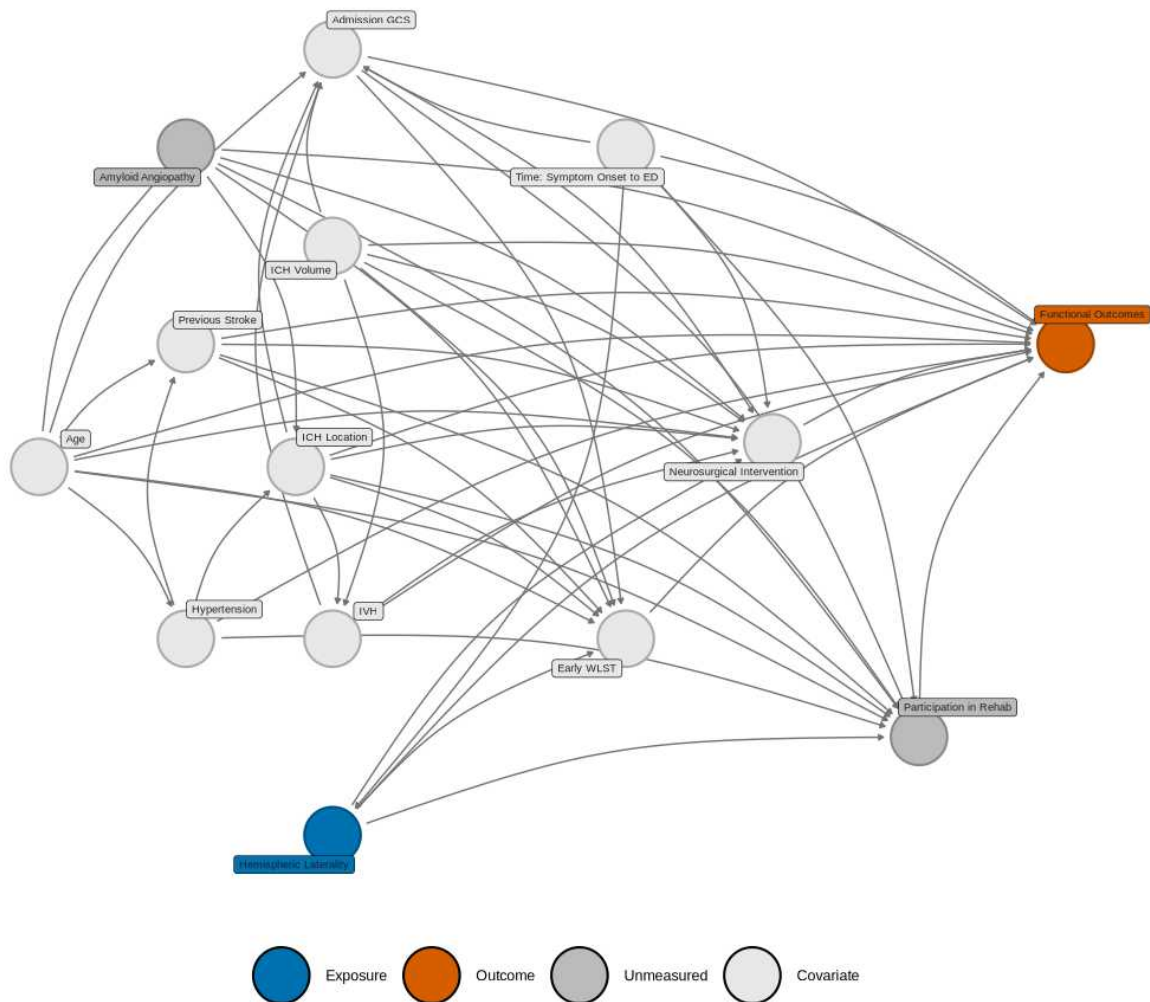
Data on midline shift, hydrocephalus, and eloquent-area involvement were not available for the analysis.

Functional Outcomes

We specified the following DAG when examining functional outcomes (e.g., modified Rankin scale score and EuroQOL measures):

Functional Outcomes

Directed acyclic graph for mRS and EuroQOL outcomes



For functional outcomes, the target estimand was likewise the total effect of hemispheric laterality on 90-day modified Rankin Scale (mRS) and EuroQOL scores. This total effect encompasses all causal pathways from laterality to functional outcome, including those mediated through neurosurgical intervention, withdrawal of life-sustaining therapy, and participation in rehabilitation. These downstream treatment pathways are intrinsic to the clinical significance of

hemispheric laterality and should not be blocked by conditioning on post-baseline treatment variables.

Conditioning on neurosurgical intervention — a mediator on the causal pathway from laterality to functional outcome — would partially block this clinically relevant pathway and could introduce collider stratification bias by opening non-causal associations between laterality and outcome within strata of the mediator. Valid estimation of a controlled direct effect (i.e., the effect of laterality holding surgical management constant) would additionally require adjustment for all confounders of the mediator–outcome relationship, some of which may themselves be affected by the exposure, creating intermediate confounding that cannot be resolved through standard regression adjustment. Accordingly, functional outcome models were adjusted for age, admission GCS score, ICH location, baseline ICH volume, and presence of IVH, but did not include neurosurgical intervention or other post-baseline treatment variables as covariates.¹

Priors

Prior Distribution Definitions

In defining prior distributions, we followed established recommendations for Bayesian modeling in critical care and clinical research.^{2–4} We defined four sets of prior distributions for the hemispheric laterality coefficient (i.e., the log-odds or log-incidence rate ratio assigned to right versus left hemisphere ICH) across all outcomes. Because the models include multiple covariates and a random study intercept, the priors describe the conditional laterality effect. The implications of these priors at the data level are characterized empirically through prior predictive checks, reported below.

- **Neutral prior:** A weakly informative prior representing the null hypothesis of no laterality effect. The coefficient is assigned a Normal(0, 0.5) distribution. This places the expected conditional OR at 1, with 95% of the prior probability mass between 0.38 and 2.66.
- **Left hemisphere prior:** An informative prior assuming outcomes are more likely following left hemisphere ICH. A Normal(-0.22, 0.175) distribution is placed on the `ich_lateralityRight` coefficient. This corresponds to an expected conditional OR of 0.8 (95% prior probability mass: 0.57 to 1.13).
- **Right hemisphere prior:** An informative prior assuming outcomes are more likely following right hemisphere ICH. A Normal(0.18, 0.175) distribution is placed on the `ich_lateralityRight` coefficient. This corresponds to an expected conditional OR of 1.2 (95% prior probability mass: 0.85 to 1.69).
- **Diffuse prior:** A vague prior intended to provide minimal information to the posterior. The coefficient is assigned a Normal(0, 5) distribution, corresponding to a conditional OR of 1 (95% prior probability mass: 0 to 1.803374^4). An exception applies to the zero-inflated negative binomial model (days of mechanical ventilation), where the diffuse prior standard deviation was restricted to 2 on the log scale to prevent prior predictive counts outside of plausible clinical ranges.

Prior Predictive Checks

Prior predictive checks were performed to understand the implications of the prior distributions before analyzing the data and ensuring the prior distributions represented plausible information regarding each outcome.

Neutral Prior

Table 1: Prior predictive check results for neutral prior — aggressive care outcomes

Outcome	Left Hemisphere (Prior expected)	Right Hemisphere (Prior expected)	Prior aOR / IRR (95% CrI)	Posterior Probability of difference (aOR > 1)	Posterior Probability of substantial difference (aOR > 1.2)	Percentage of posterior within ROPE
Neurosurgical Intervention	20.9% (1.8 - 86.1%)	22.0% (2.4 - 85.3%)	0.99 (0.68 - 1.49)	48%	18%	30%
External Ventricular Drain	50.6% (32.7 - 83.1%)	49.7% (34.1 - 84.4%)	0.99 (0.77 - 1.29)	46%	6%	46%
Days of Mechanical Ventilation	8.3 (3 - 36.7)	8.3 (3.1 - 36.3)	0.99 (0.91 - 1.08)	42%	< 0.1%	60%
Tracheostomy	5.1% (0.0 - 24.6%)	5.7% (0.0 - 25.0%)	1 (0.49 - 1.72)	48%	12%	42%
Withdrawal of Life-Sustaining Therapy	18.3% (0.8 - 47.8%)	18.9% (0.7 - 47.8%)	0.99 (0.49 - 1.32)	42%	12%	40%
Early Withdrawal of Life-Sustaining Therapy	50.6% (32.7 - 83.1%)	49.7% (34.1 - 84.4%)	0.99 (0.77 - 1.29)	46%	6%	46%
DNR Order	18.3% (0.8 - 47.8%)	18.9% (0.7 - 47.8%)	0.99 (0.49 - 1.32)	42%	12%	40%

Values represent expected values simulated purely from the prior distributions before encountering the data. ROPE indicates region of practical equivalence (0.95 to 1.05); aOR, adjusted odds ratio; IRR, incidence rate ratio; and CrI, credible interval.

Table 2: Prior predictive check results for neutral prior — functional outcomes

Outcome	Prior Effect Estimate (95% CrI)	Prior Prob. of any diff	Prior Prob. of substantial diff	Percentage of prior within ROPE
Modified Rankin Score	1.00 (0.83 - 1.28)	46%	4%	66%
EuroQOL - Mobility	1.01 (0.93 - 1.17)	60%	2%	72%
EuroQOL - Self-Care	1.01 (0.93 - 1.17)	60%	2%	72%
EuroQOL - Usual Activities	1.01 (0.93 - 1.17)	60%	2%	72%
EuroQOL - Pain/Discomfort	1.01 (0.95 - 1.12)	66%	2%	74%
EuroQOL - Anxiety/Depression	1.01 (0.95 - 1.12)	64%	2%	74%
EQ-VAS	-3.8 (-25.8 - 24.0)	60%	44%	6%

Values represent expected effects simulated purely from the prior distributions before encountering the data. Estimates are Prior Odds Ratios (aOR) for ordinal outcomes and Mean Differences for EQ-VAS. ROPE indicates region of practical equivalence (0.95 to 1.05 for aOR; ± 2 points for EQ-VAS).

Left Hemisphere Prior

Table 3: Prior predictive check results for left hemisphere prior — aggressive care outcomes

Outcome	Left Hemisphere (Prior expected)	Right Hemisphere (Prior expected)	Prior aOR / IRR (95% CrI)	Posterior Probability of difference (aOR > 1)	Posterior Probability of substantial difference (aOR > 1.2)	Percentage of posterior within ROPE
Neurosurgical Intervention	21.2% (1.7 - 56.6%)	21.2% (2.1 - 51.9%)	0.98 (0.78 - 1.14)	36%	2%	62%
External Ventricular Drain	54.9% (29.5 - 81.0%)	52.1% (30.9 - 74.8%)	0.97 (0.79 - 1.12)	22%	2%	74%
Days of Mechanical Ventilation	4.3 (0.9 - 8.3)	3.4 (0.7 - 6.5)	0.83 (0.62 - 1.11)	8%	2%	16%
Tracheostomy	6.6% (0.0 - 51.6%)	6.9% (0.0 - 48.2%)	0.97 (0.69 - 1.04)	18%	2%	52%
Withdrawal of Life-Sustaining Therapy	26.2% (0.4 - 55.4%)	26.9% (0.6 - 54.1%)	0.96 (0.69 - 1.03)	14%	< 0.1%	56%
Early Withdrawal of Life-Sustaining Therapy	54.9% (29.5 - 81.0%)	52.1% (30.9 - 74.8%)	0.97 (0.79 - 1.12)	22%	2%	74%
DNR Order	26.2% (0.4 - 55.4%)	26.9% (0.6 - 54.1%)	0.96 (0.69 - 1.03)	14%	< 0.1%	56%

Values represent expected values simulated purely from the prior distributions before encountering the data. ROPE indicates region of practical equivalence (0.95 to 1.05); aOR, adjusted odds ratio; IRR, incidence rate ratio; and CrI, credible interval.

Table 4: Prior predictive check results for left hemisphere prior — functional outcomes

Outcome	Prior Effect Estimate (95% CrI)	Prior Prob. of any diff	Prior Prob. of substantial diff	Percentage of prior within ROPE
Modified Rankin Score	1.00 (0.85 - 1.16)	58%	2%	80%
EuroQOL - Mobility	1.00 (0.94 - 1.25)	54%	8%	82%
EuroQOL - Self-Care	1.00 (0.94 - 1.25)	54%	8%	82%
EuroQOL - Usual Activities	1.00 (0.94 - 1.25)	54%	8%	82%
EuroQOL - Pain/Discomfort	1.00 (0.96 - 1.22)	54%	4%	90%
EuroQOL - Anxiety/Depression	1.00 (0.96 - 1.22)	54%	4%	86%
EQ-VAS	1.5 (-20.2 - 14.8)	46%	28%	22%

Values represent expected effects simulated purely from the prior distributions before encountering the data. Estimates are Prior Odds Ratios (aOR) for ordinal outcomes and Mean Differences for EQ-VAS. ROPE indicates region of practical equivalence (0.95 to 1.05 for aOR; ± 2 points for EQ-VAS).

Right Hemisphere Prior

Table 5: Prior predictive check results for right hemisphere prior — aggressive care outcomes

Outcome	Left Hemisphere (Prior expected)	Right Hemisphere (Prior expected)	Prior aOR / IRR (95% CrI)	Posterior Probability of difference (aOR > 1)	Posterior Probability of substantial difference (aOR > 1.2)	Percentage of posterior within ROPE
Neurosurgical Intervention	20.3% (3.9 - 49.9%)	22.8% (3.4 - 48.2%)	1.03 (0.96 - 1.21)	76%	4%	62%
External Ventricular Drain	45.0% (15.7 - 68.3%)	47.2% (14.3 - 68.1%)	1.02 (0.89 - 1.11)	70%	2%	64%
Days of Mechanical Ventilation	5.6 (1.5 - 14.8)	6.2 (1.8 - 20.9)	1.16 (0.98 - 1.49)	92%	46%	16%
Tracheostomy	5.3% (0.0 - 24.6%)	6.1% (0.0 - 25.6%)	1.05 (0.88 - 2.23)	74%	18%	52%
Withdrawal of Life-Sustaining Therapy	25.5% (2.8 - 50.3%)	26.6% (3.7 - 52.0%)	1.03 (0.93 - 2.06)	78%	12%	56%
Early Withdrawal of Life-Sustaining Therapy	45.0% (15.7 - 68.3%)	47.2% (14.3 - 68.1%)	1.02 (0.89 - 1.11)	70%	2%	64%
DNR Order	25.5% (2.8 - 50.3%)	26.6% (3.7 - 52.0%)	1.03 (0.93 - 2.06)	78%	12%	56%

Values represent expected values simulated purely from the prior distributions before encountering the data. ROPE indicates region of practical equivalence (0.95 to 1.05); aOR, adjusted odds ratio; IRR, incidence rate ratio; and CrI, credible interval.

Table 6: Prior predictive check results for right hemisphere prior — functional outcomes

Outcome	Prior Effect Estimate (95% CrI)	Prior Prob. of any diff	Prior Prob. of substantial diff	Percentage of prior within ROPE
Modified Rankin Score	1.00 (0.83 - 1.27)	40%	4%	68%
EuroQOL - Mobility	1.00 (0.86 - 1.08)	46%	2%	86%
EuroQOL - Self-Care	1.00 (0.86 - 1.08)	46%	2%	86%
EuroQOL - Usual Activities	1.00 (0.86 - 1.08)	46%	2%	86%
EuroQOL - Pain/Discomfort	1.00 (0.92 - 1.06)	50%	2%	86%
EuroQOL - Anxiety/Depression	1.00 (0.92 - 1.06)	52%	2%	88%
EQ-VAS	1.3 (-20.8 - 22.3)	46%	34%	14%

Values represent expected effects simulated purely from the prior distributions before encountering the data. Estimates are Prior Odds Ratios (aOR) for ordinal outcomes and Mean Differences for EQ-VAS. ROPE indicates region of practical equivalence (0.95 to 1.05 for aOR; ± 2 points for EQ-VAS).

Diffuse Prior

Table 7: Prior predictive check results for diffuse prior — aggressive care outcomes

Outcome	Left Hemisphere (Prior expected)	Right Hemisphere (Prior expected)	Prior aOR / IRR (95% CrI)	Posterior Probability of difference (aOR > 1)	Posterior Probability of substantial difference (aOR > 1.2)	Percentage of posterior within ROPE
Neurosurgical Intervention	48.8% (27.8 - 68.4%)	47.8% (30.0 - 66.6%)	0.99 (0.87 - 1.15)	46%	< 0.1%	56%
External Ventricular Drain	49.8% (30.9 - 70.7%)	49.8% (33.4 - 67.3%)	1.03 (0.81 - 1.3)	58%	8%	28%
Days of Mechanical Ventilation	3212.2 (37.3 - 521285)	1830.2 (55.4 - 686812.3)	1 (0.63 - 1.63)	50%	22%	26%
Tracheostomy	40.0% (21.9 - 61.6%)	39.6% (23.7 - 62.0%)	0.99 (0.79 - 1.25)	46%	8%	28%
Withdrawal of Life-Sustaining Therapy	43.0% (29.1 - 68.3%)	45.1% (30.6 - 62.8%)	1 (0.75 - 1.19)	50%	2%	28%
Early Withdrawal of Life-Sustaining Therapy	49.8% (30.9 - 70.7%)	49.8% (33.4 - 67.3%)	1.03 (0.81 - 1.3)	58%	8%	28%
DNR Order	43.0% (29.1 - 68.3%)	45.1% (30.6 - 62.8%)	1 (0.75 - 1.19)	50%	2%	28%

Values represent expected values simulated purely from the prior distributions before encountering the data. ROPE indicates region of practical equivalence (0.95 to 1.05); aOR, adjusted odds ratio; IRR, incidence rate ratio; and CrI, credible interval.

Table 8: Prior predictive check results for diffuse prior — functional outcomes

Outcome	Prior Effect Estimate (95% CrI)	Prior Prob. of any diff	Prior Prob. of substantial diff	Percentage of prior within ROPE
Modified Rankin Score	1.02 (0.01 - 136.98)	50%	40%	6%
EuroQOL - Mobility	0.96 (0.26 - 7.81)	42%	32%	22%
EuroQOL - Self-Care	0.96 (0.26 - 7.81)	42%	32%	22%
EuroQOL - Usual Activities	0.96 (0.26 - 7.81)	42%	32%	22%
EuroQOL - Pain/Discomfort	1.01 (0.11 - 8.09)	50%	34%	16%
EuroQOL - Anxiety/Depression	1.04 (0.10 - 8.47)	56%	30%	14%
EQ-VAS	0.7 (-27.9 - 33.8)	48%	26%	22%

Values represent expected effects simulated purely from the prior distributions before encountering the data. Estimates are Prior Odds Ratios (aOR) for ordinal outcomes and Mean Differences for EQ-VAS. ROPE indicates region of practical equivalence (0.95 to 1.05 for aOR; ± 2 points for EQ-VAS).

Missing Data Assessment

As described in the primary manuscript, the primary analysis used multiple imputation by chained equations (MICE) under a Missing at Random (MAR) assumption. This section provides detailed reporting of missingness patterns across the analytic cohort, assessments supporting the MAR assumption, and diagnostics evaluating the quality of the imputed datasets.

Missingness for 90-Day Functional Outcomes by Laterality Group

Baseline covariates used in model adjustment (age, GCS, ICH volume, ICH location, IVH) had minimal missingness (<1%). However, 90-day follow-up outcomes had more substantial missingness due to loss to follow-up and differences in outcome ascertainment between the ATACH-2 and ERICH study protocols. The table below summarizes missingness for each 90-day outcome by hemispheric laterality group.

Table 9: Missingness for 90-Day Outcomes by Hemispheric Laterality

Outcome	Left Hemisphere	Right Hemisphere
EQ-5D Anxiety/Depression	477/1720 (27.7%)	468/1757 (26.6%)
EQ-5D Mobility	463/1720 (26.9%)	455/1757 (25.9%)
EQ-5D Pain/Discomfort	470/1720 (27.3%)	466/1757 (26.5%)
EQ-5D Self-Care	463/1720 (26.9%)	455/1757 (25.9%)
EQ-5D Usual Activities	463/1720 (26.9%)	455/1757 (25.9%)
EQ-VAS at 90 Days	606/1720 (35.2%)	614/1757 (34.9%)
mRS at 90 Days	210/1720 (12.2%)	212/1757 (12.1%)

Values are n missing / N total (%). mRS indicates modified Rankin Scale; EQ-VAS, European Quality of Life Visual Analog Scale; EQ-5D, European Quality of Life-5 Dimensions.

Missingness Patterns by Laterality and Study

The following figures show the percentage of missing data across all analysis variables, stratified by hemispheric laterality and by study. Missingness patterns were largely driven by differences between the ATACH-2 and ERIICH study protocols rather than by hemispheric laterality.

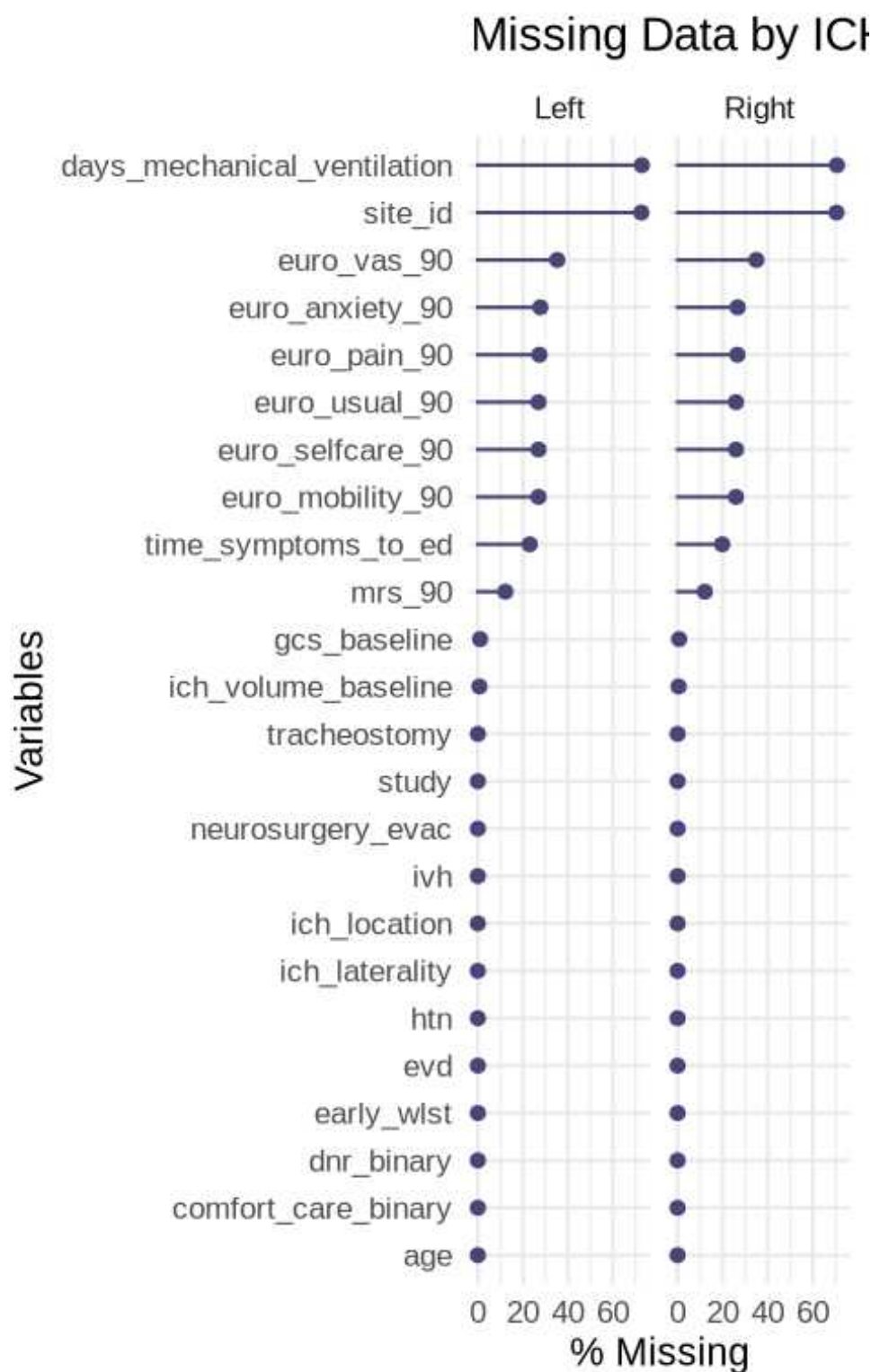


Figure 1: Percentage of missing data by hemispheric laterality

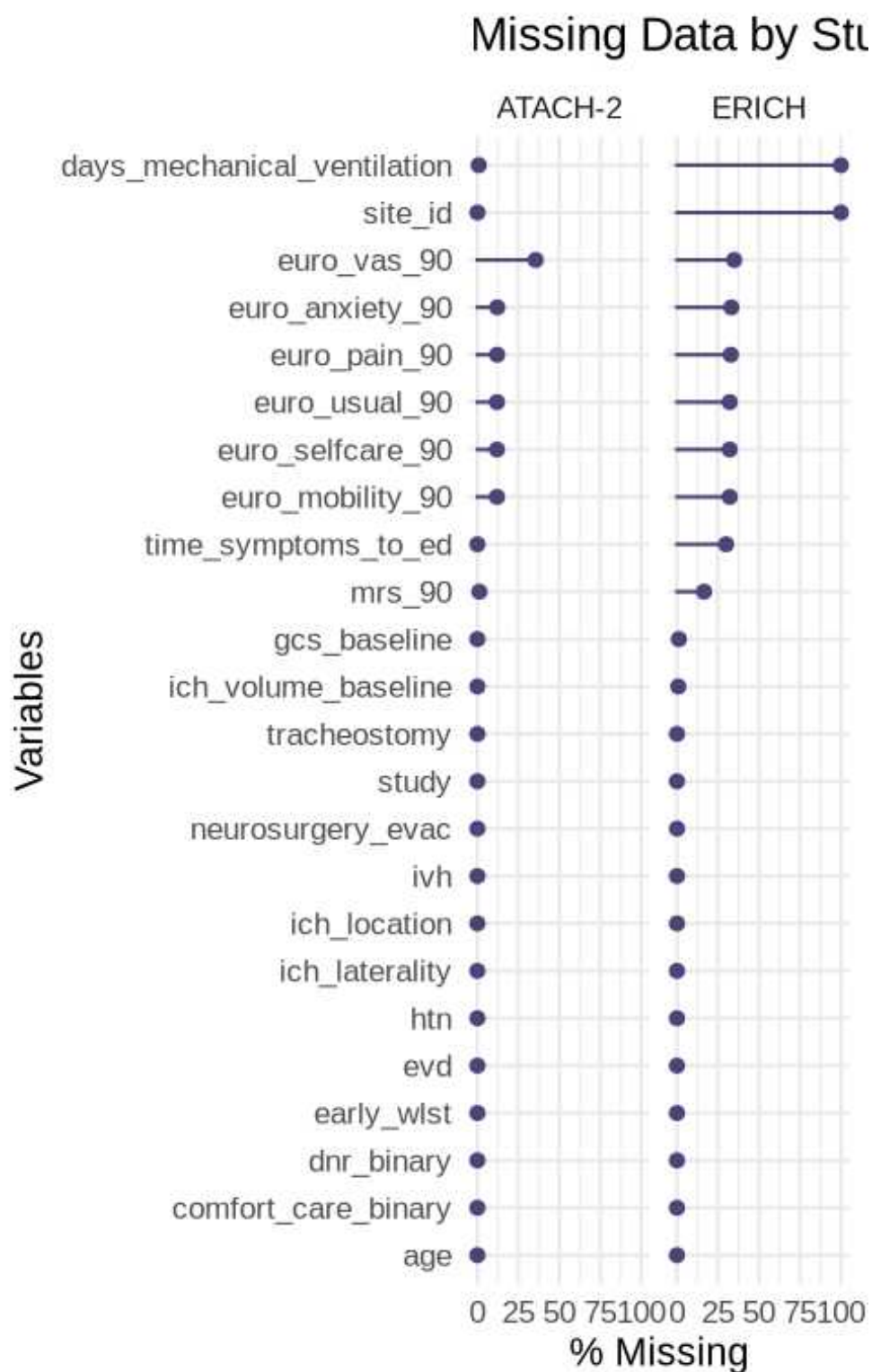
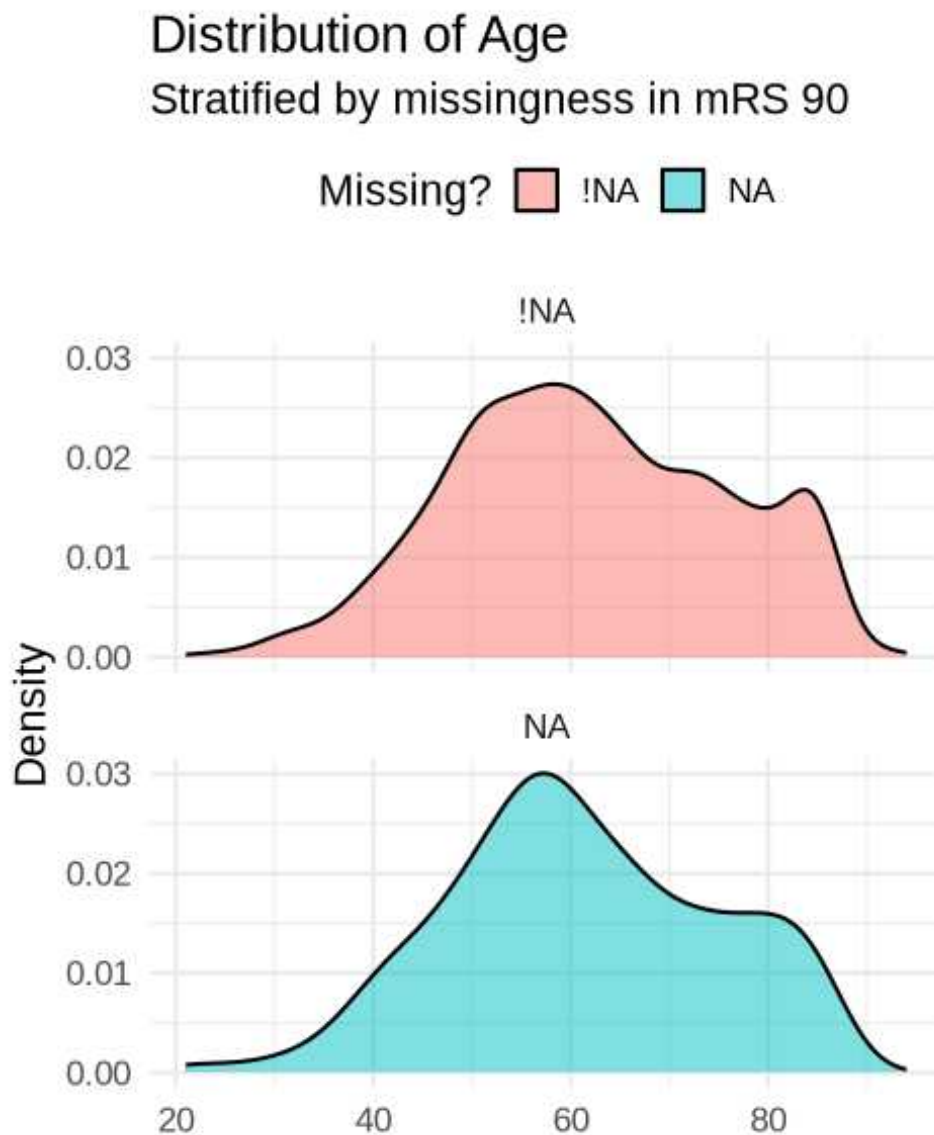


Figure 2: Percentage of missing data by study

Assessment of Missing at Random Assumption

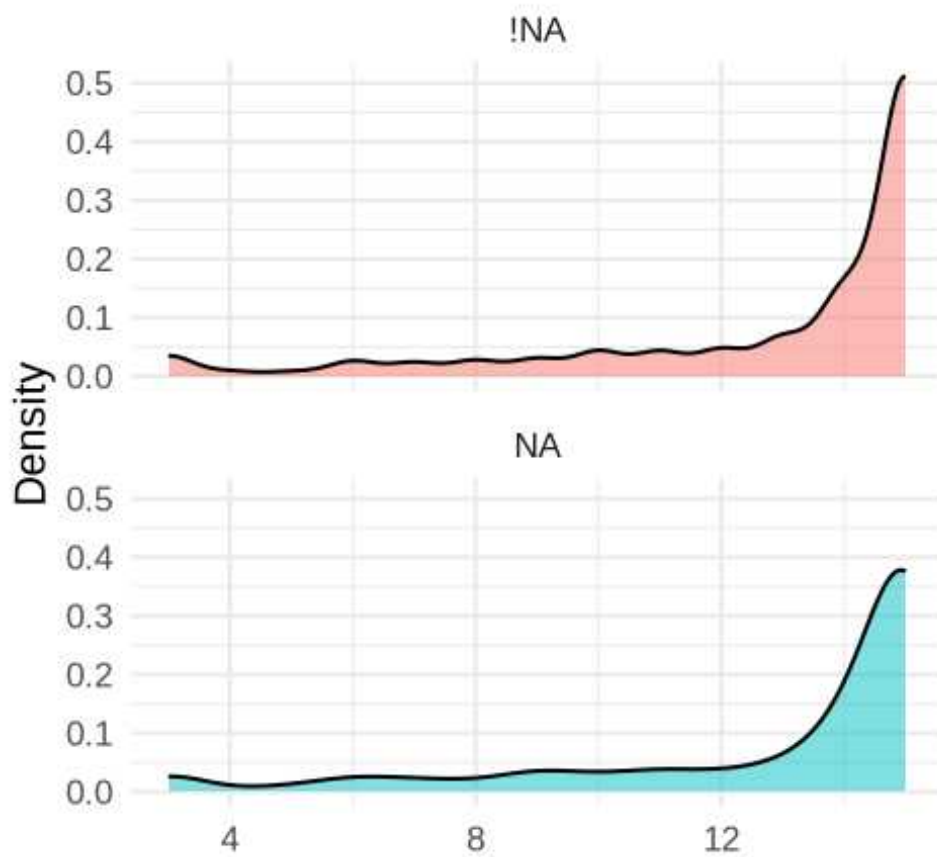
To assess whether missingness in 90-day outcomes was associated with key baseline covariates, we generated shadow plots comparing the distribution of each covariate among patients with observed versus missing values. If the distributions are similar, this supports the MAR assumption — that missingness is conditionally independent of the unobserved outcome given the observed covariates.

mRS at 90 Days



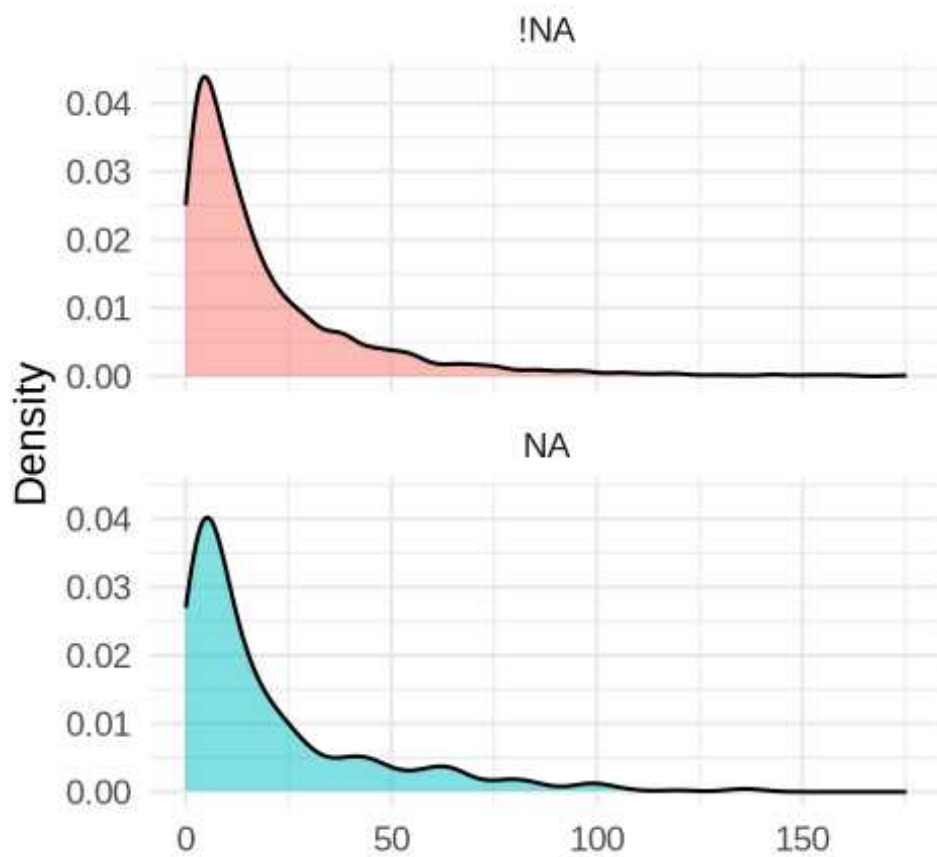
Distribution of GCS Baseline Stratified by missingness in mRS 90

Missing? ■ !NA ■ NA



Distribution of ICH Volume Baseline Stratified by missingness in mRS 90

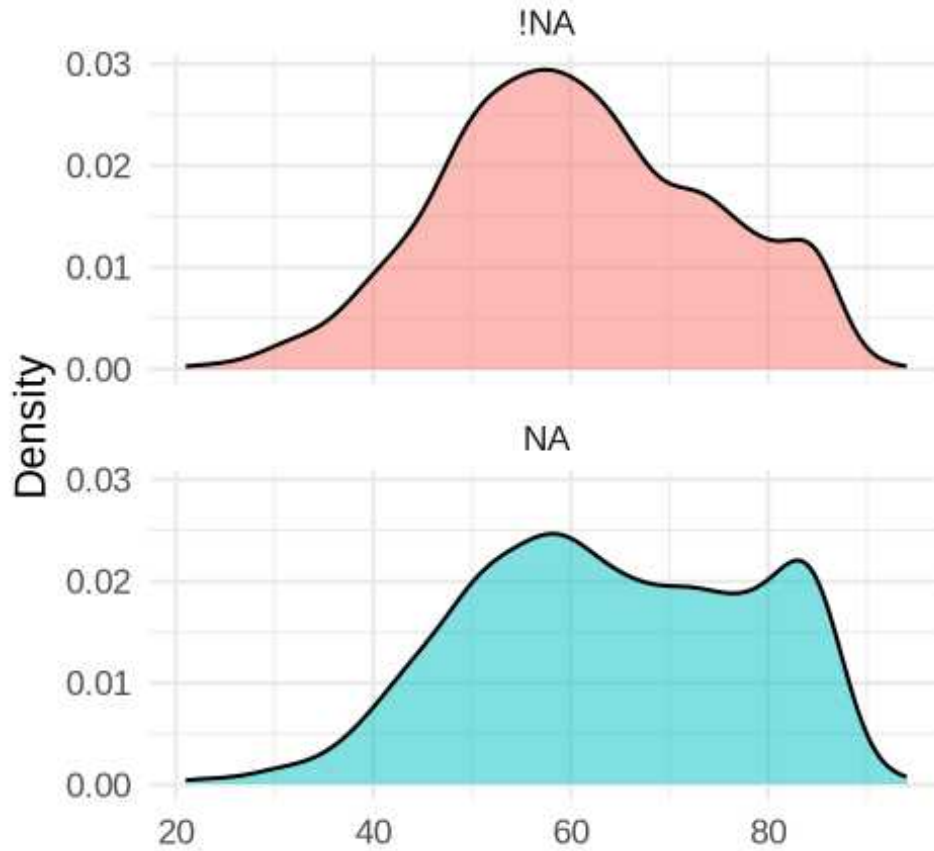
Missing? ■ !NA ■ NA



Distribution of Age

Stratified by missingness in Euro VAS 90

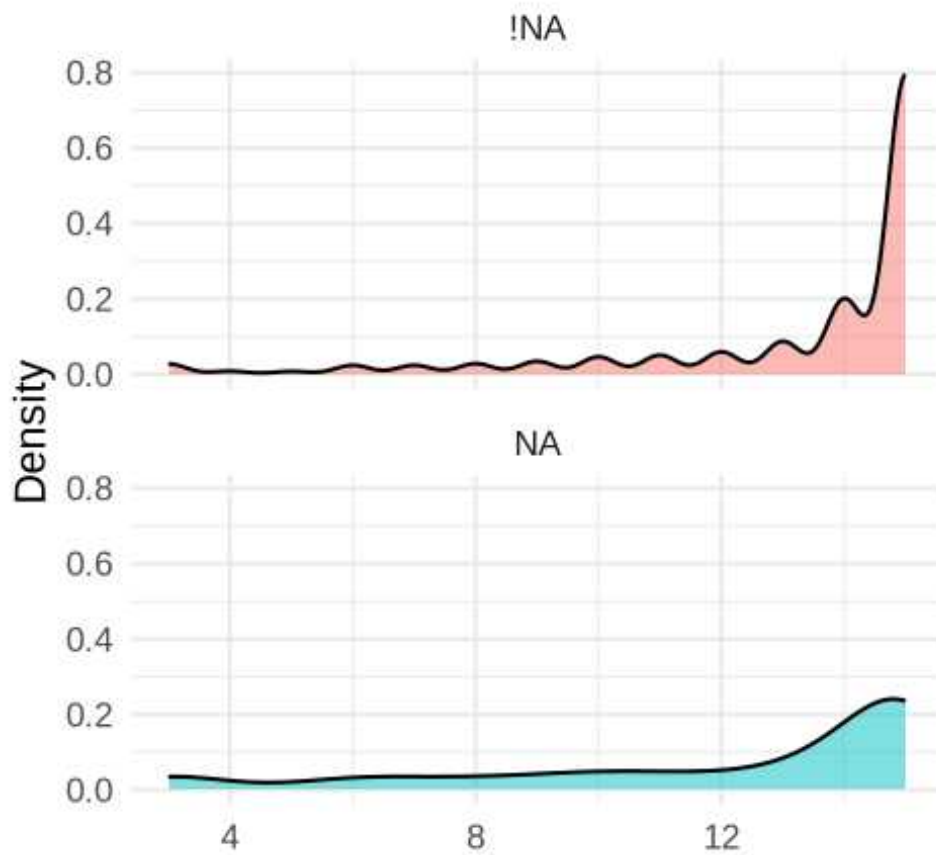
Missing? ■ !NA ■ NA



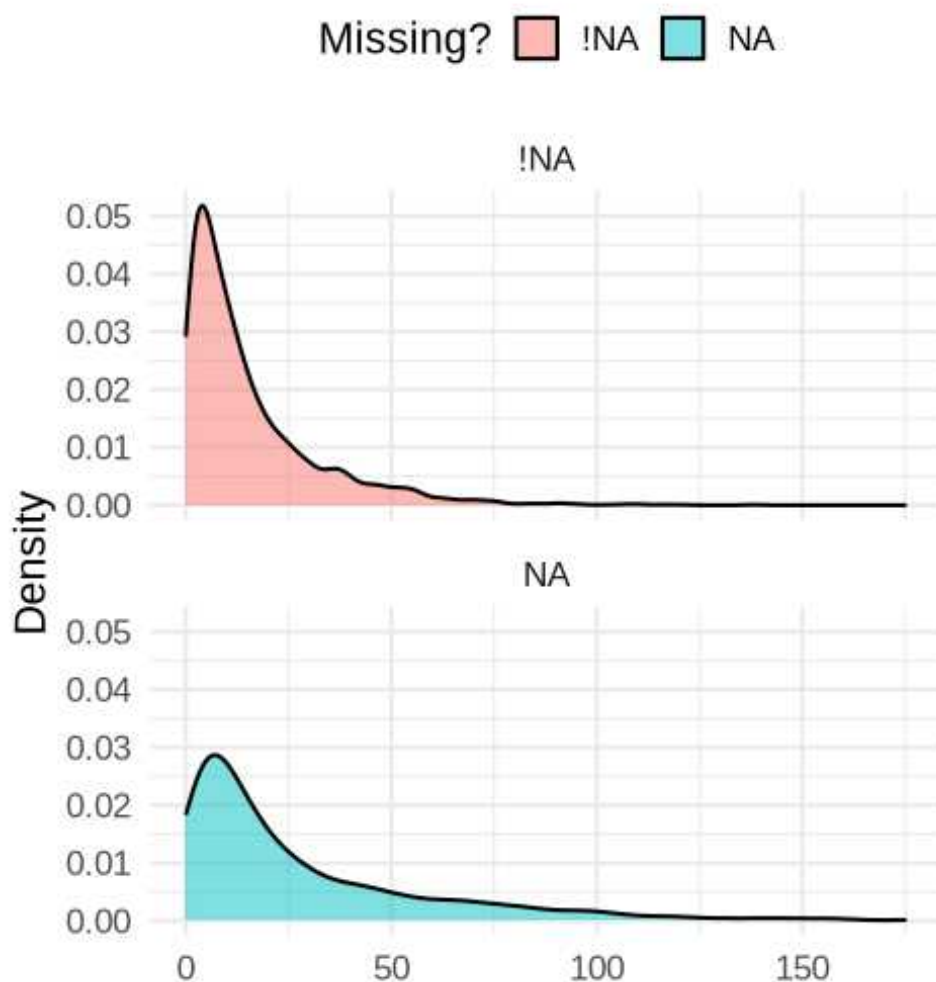
Distribution of GCS Baseline

Stratified by missingness in Euro VAS 90

Missing? ■ !NA ■ NA



Distribution of ICH Volume Baseline Stratified by missingness in Euro VAS 90

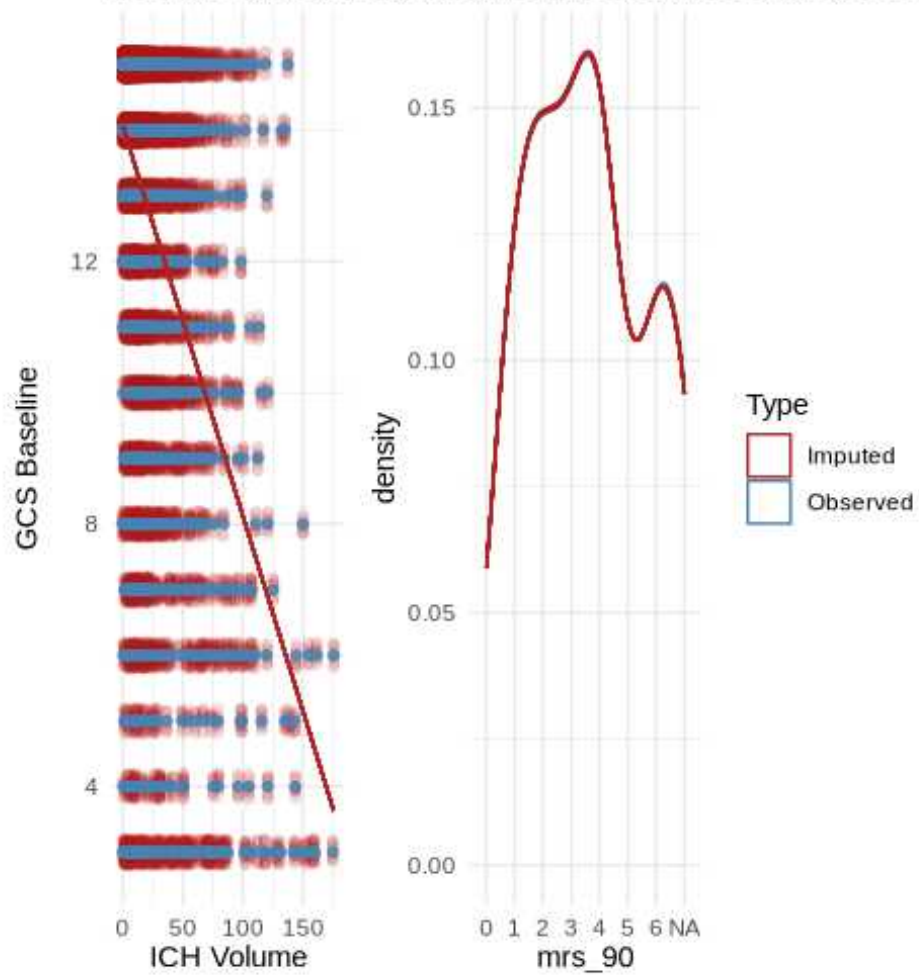


Imputation Diagnostics

The following figure presents diagnostic checks for the MICE imputation procedure, comparing the distributions of observed data against imputed values across the 20 imputed datasets.

Consistency between the observed and imputed distributions supports the plausibility of the imputation model.

Bivariate Consistency: Volume vs GCS
 Outcome: mRS 90
 Red lines (Imputations) should bundle around Blue line (Observed)



Type — Imputed — Observed

Categorical Distribution: ICH Location



Posterior Simulation Diagnostics

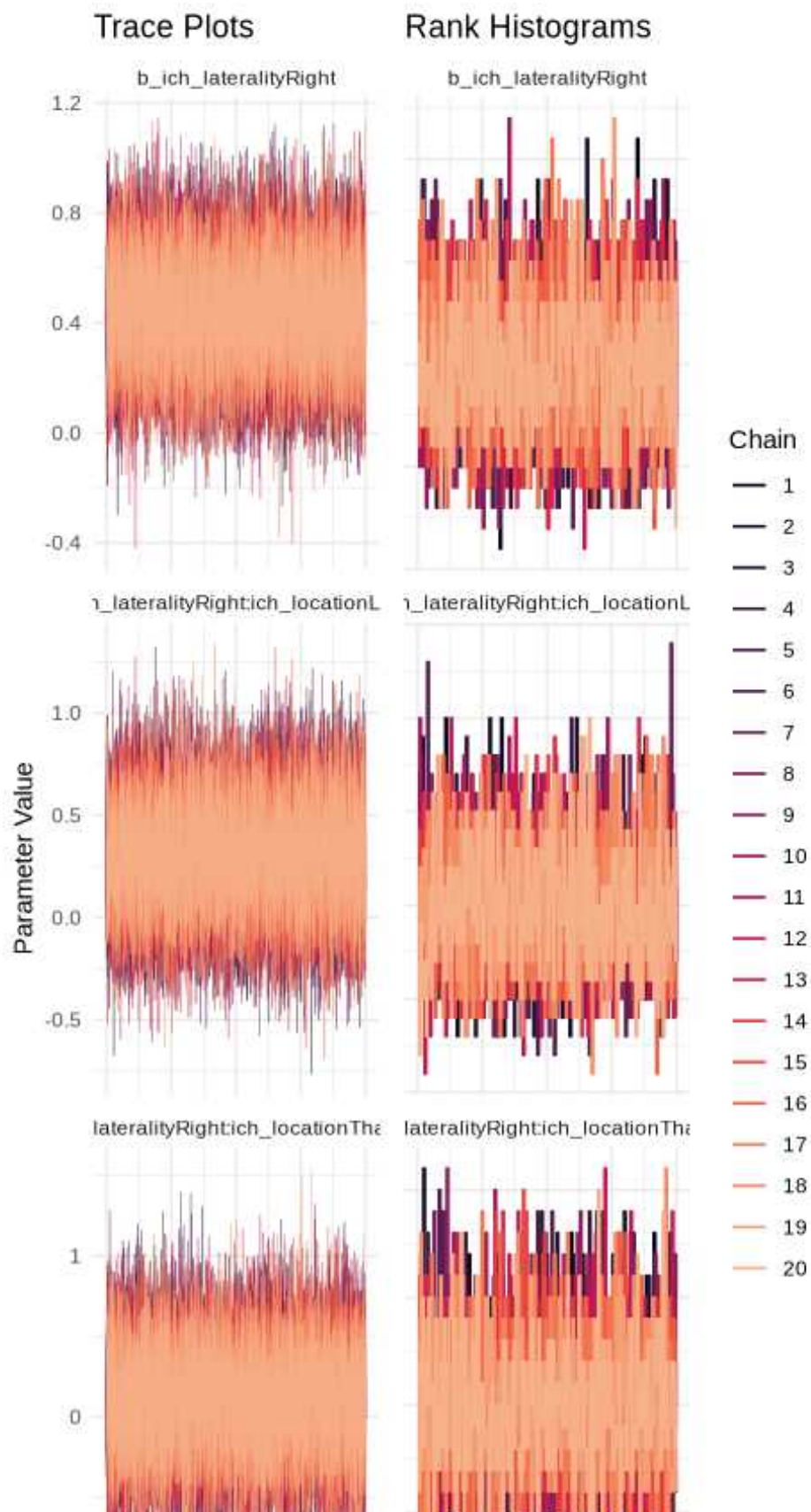
Posterior Simulation Checks

To ensure the reliability and validity of our Bayesian models, we performed several posterior simulation checks.

Chain Convergence: We assessed the convergence of the Markov Chain Monte Carlo (MCMC) chains by visually inspecting trace plots for each parameter. The trace plots showed that the chains were well-mixed and stationary, indicating that they had converged to a stable posterior distribution. Additionally, we calculated the R-hat statistic for all parameters, ensuring that all values were below 1.01, which confirms the convergence of the chains.

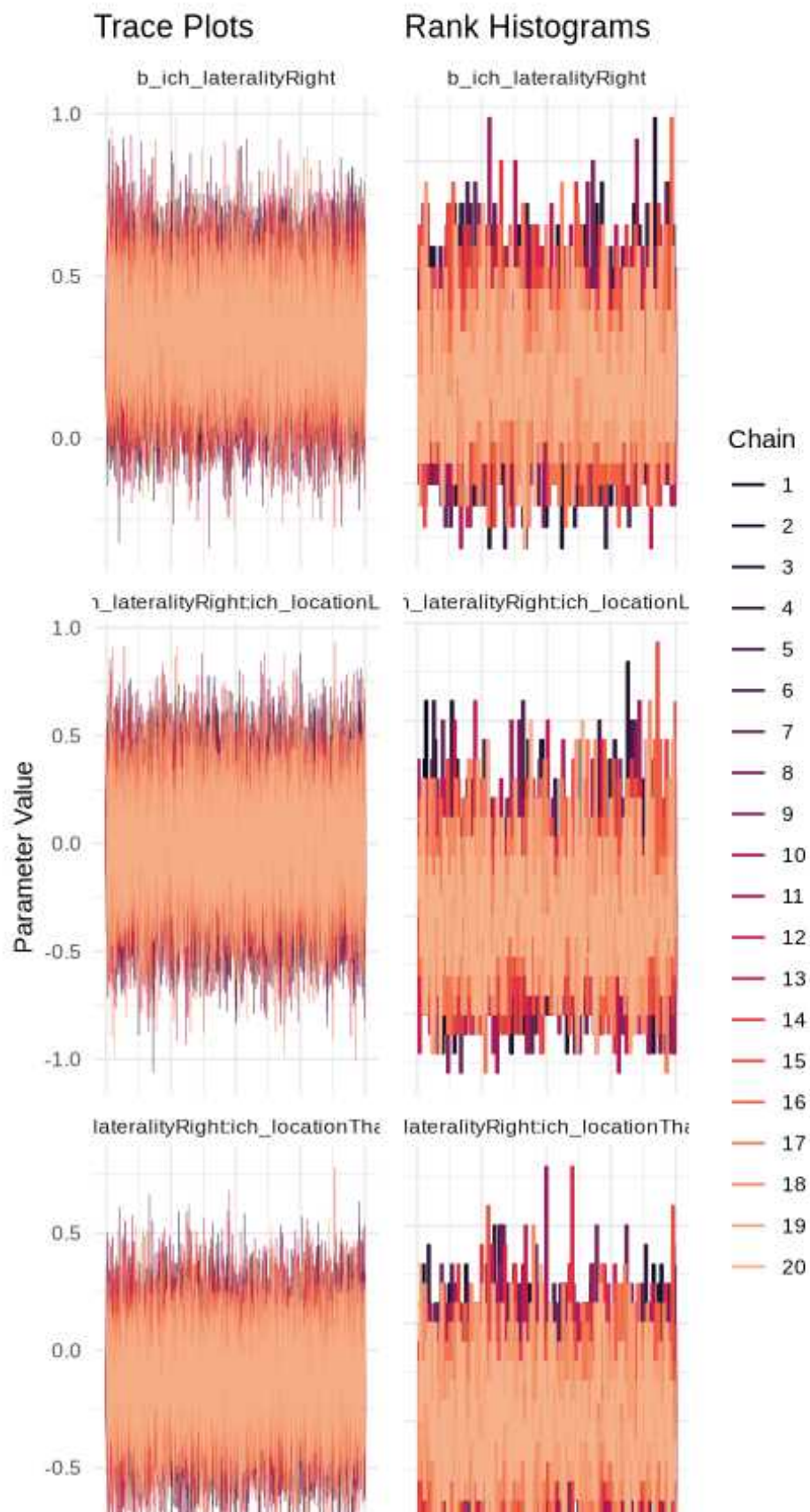
Posterior Predictive Checks: We conducted posterior predictive checks to evaluate the model's fit to the data. This involved simulating data from the posterior predictive distribution and comparing it to the observed data. The close correspondence between the simulated and observed data confirmed that our models provided a good fit to the underlying data structure.

MCMC Chain Convergence and Resolution
Neurosurgical Intervention



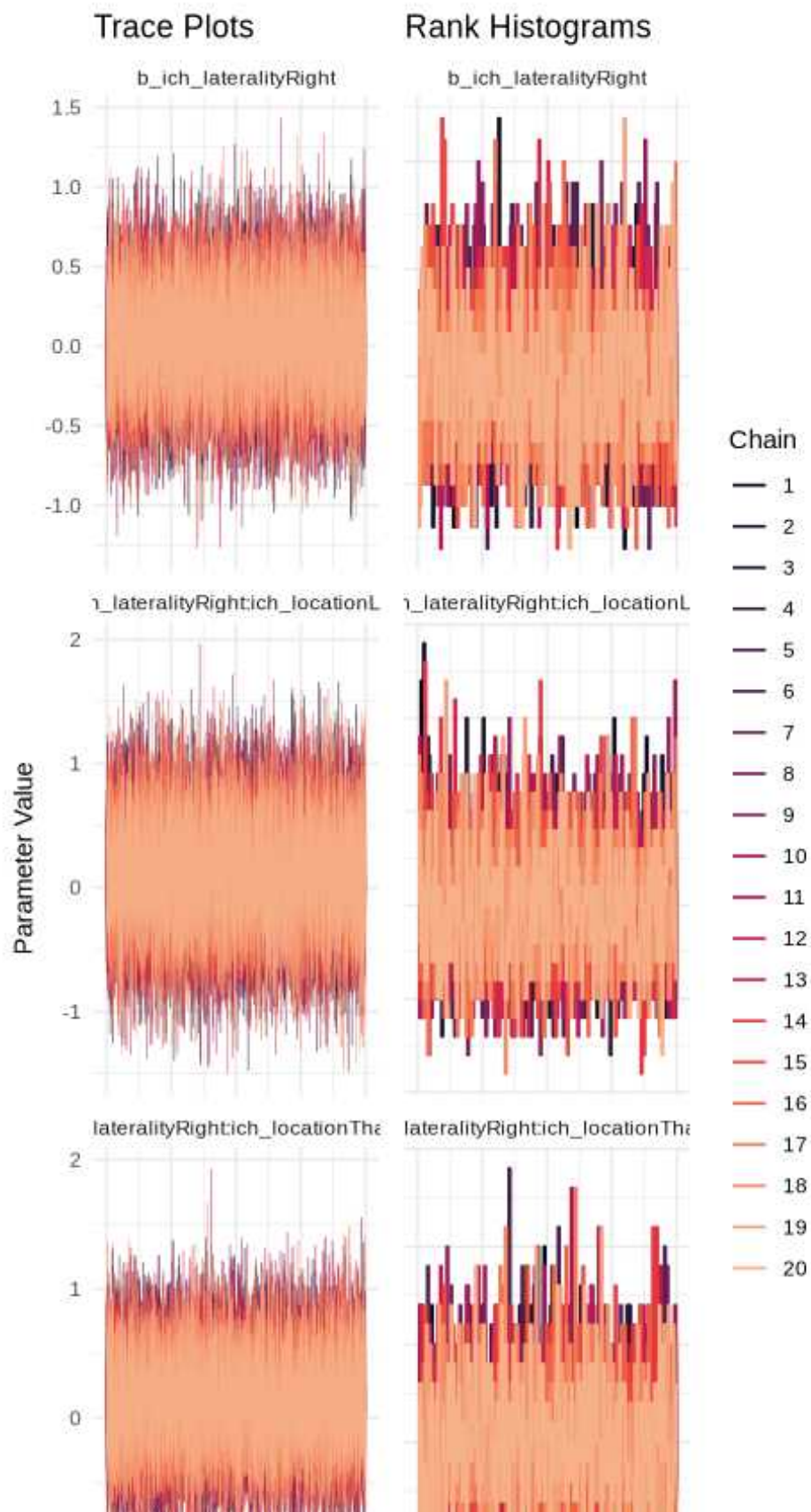
Posterior diagnostic checks for Neurosurgical Intervention

Ventriculostomy



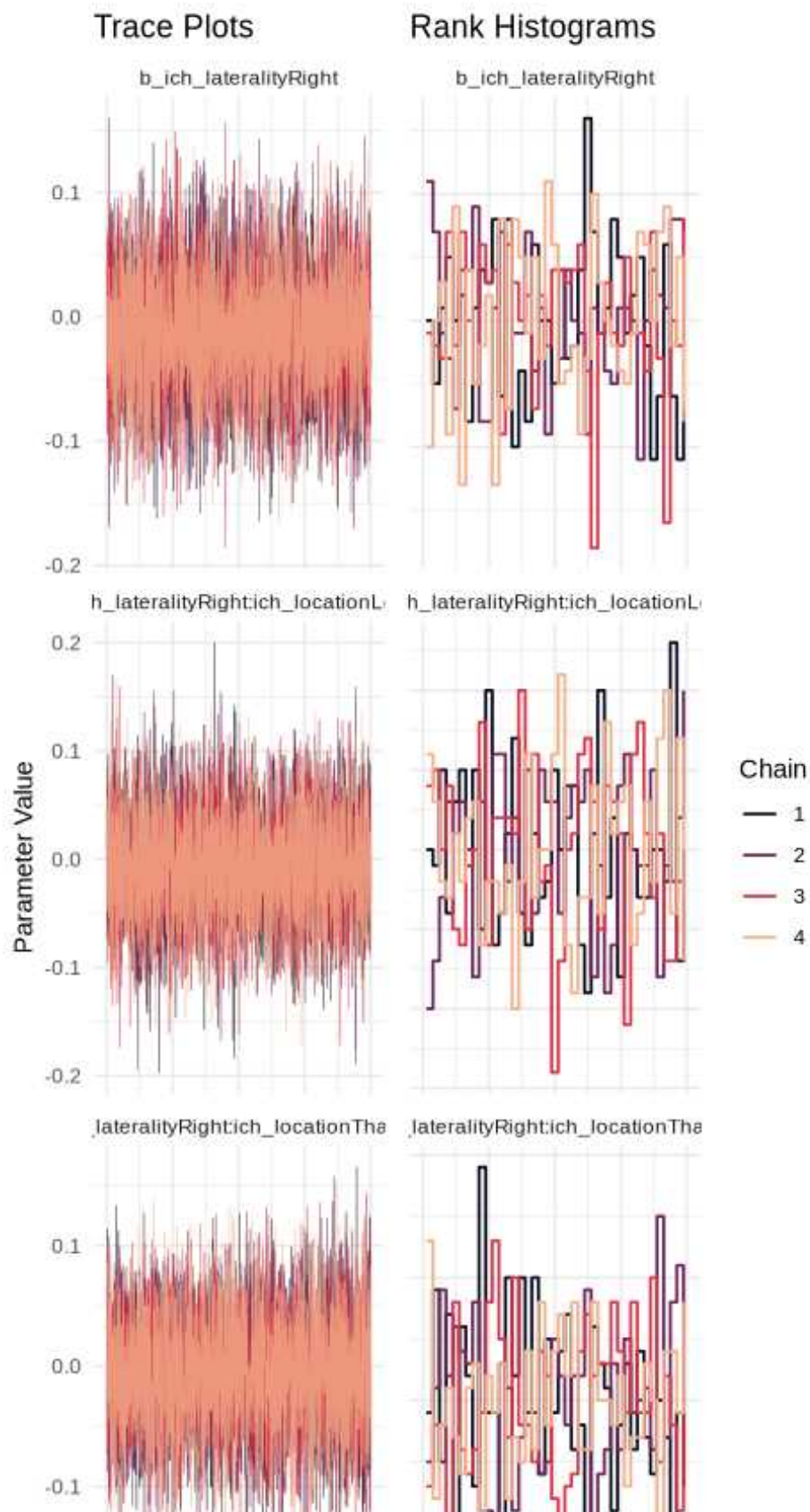
Posterior diagnostic checks for Ventriculostomy

Tracheostomy



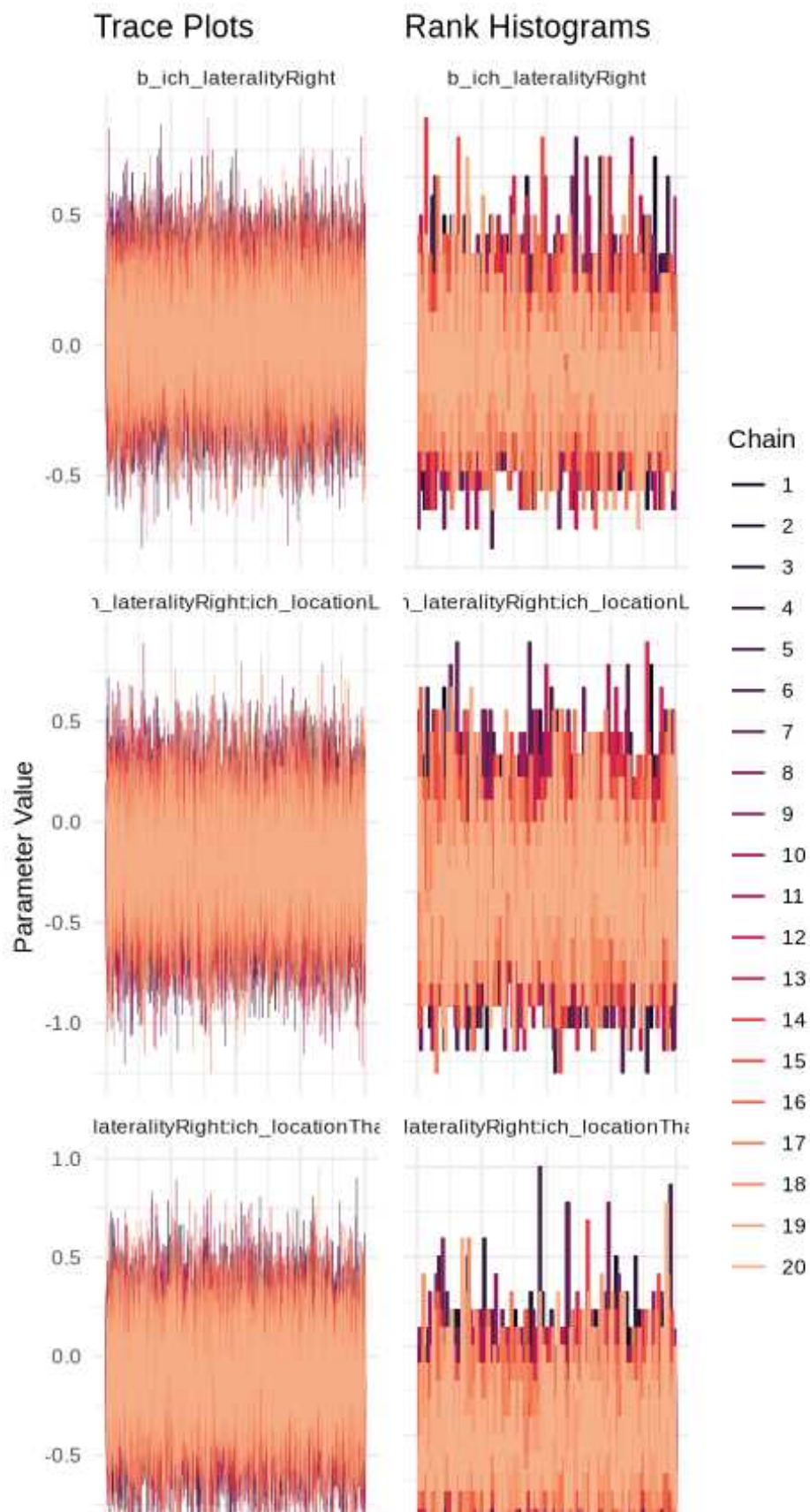
Posterior diagnostic checks for Tracheostomy

Days of Mechanical Ventilation



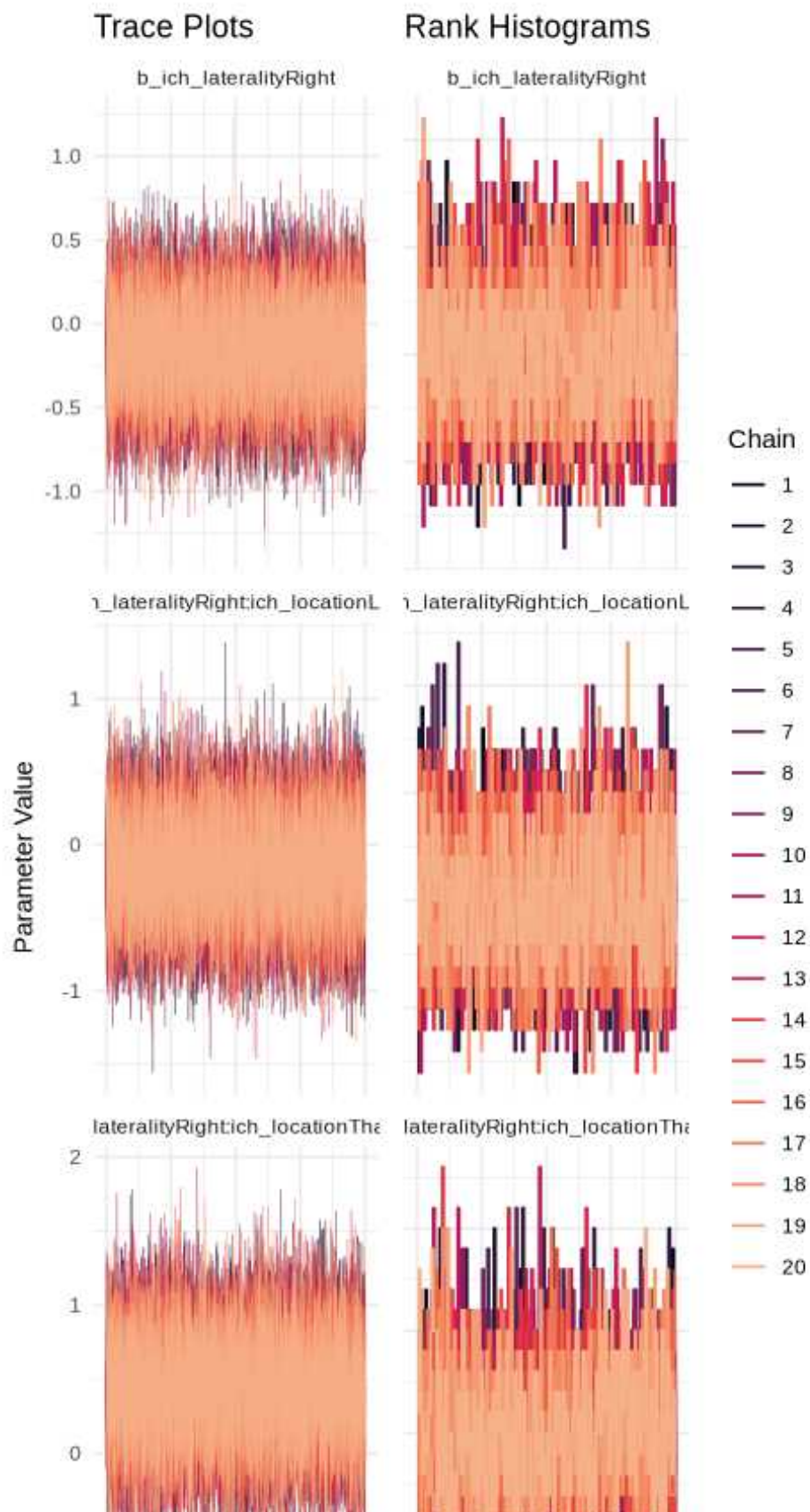
Posterior diagnostic checks for Days of Mechanical Ventilation

WLST



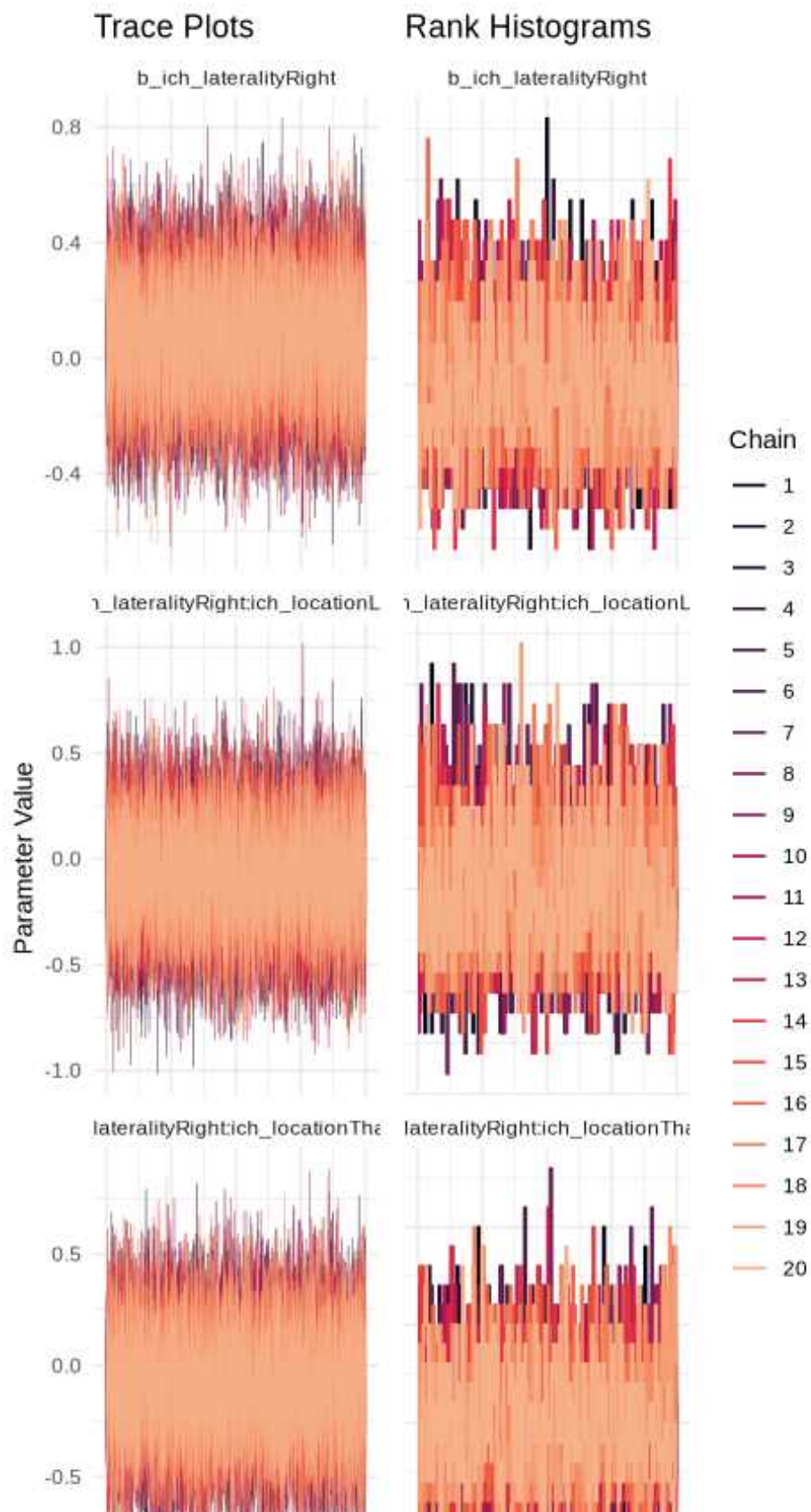
Posterior diagnostic checks for WLST

Early WLST



Posterior diagnostic checks for Early WLST

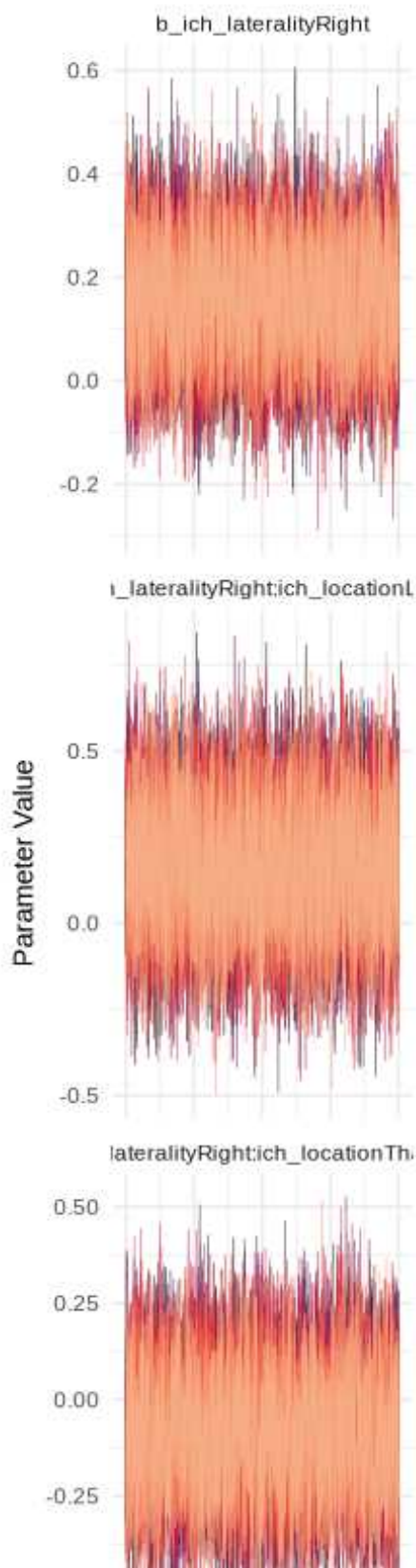
DNR



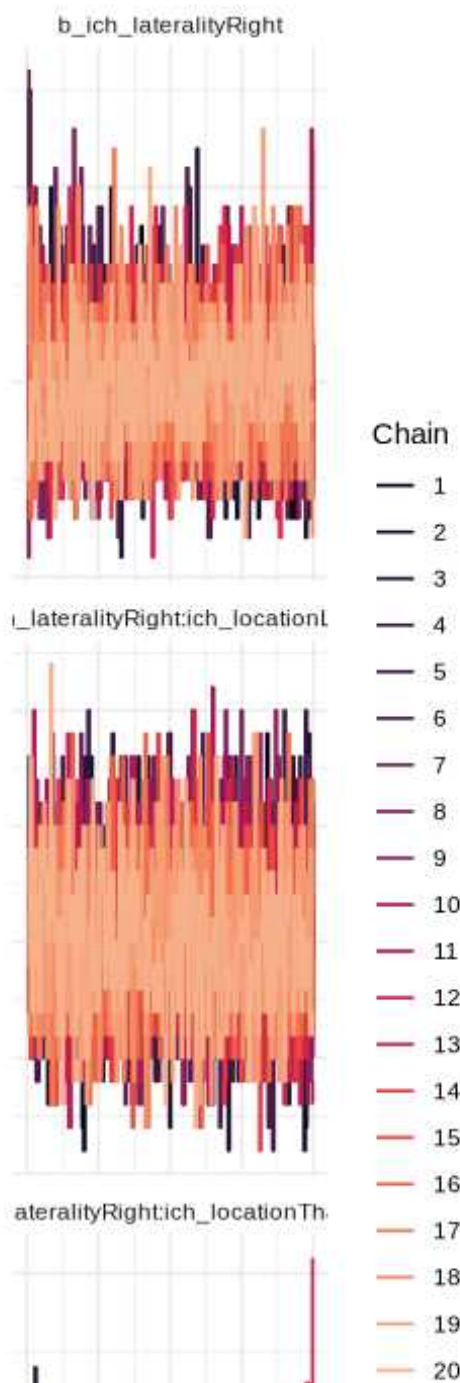
Posterior diagnostic checks for DNR

mRS 90

Trace Plots



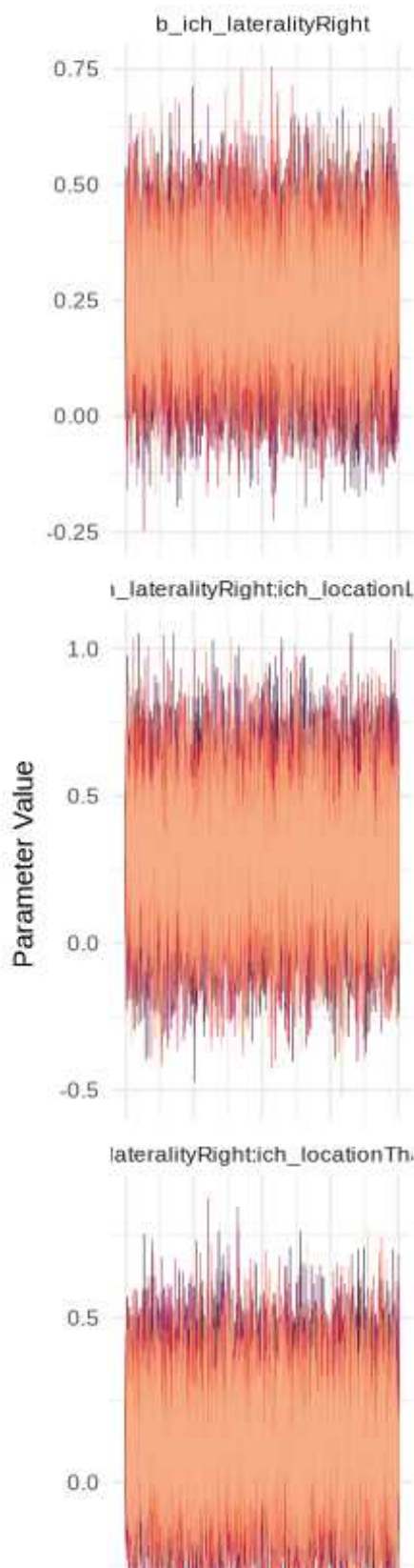
Rank Histograms



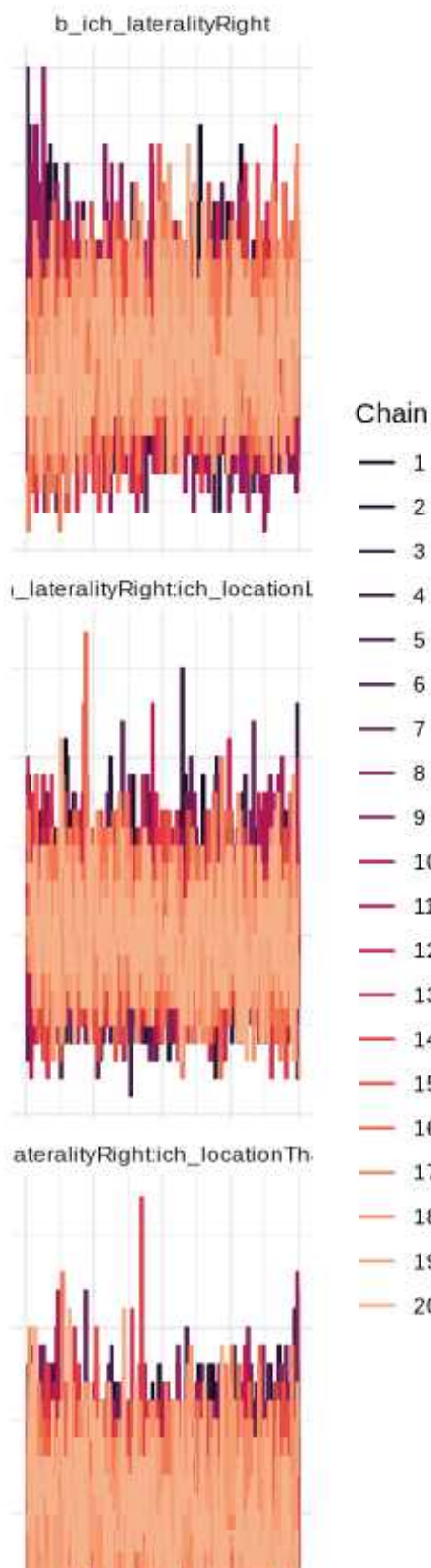
Posterior diagnostic checks for Modified Rankin Scale at 90 Days

EuroQOL Mobility 90

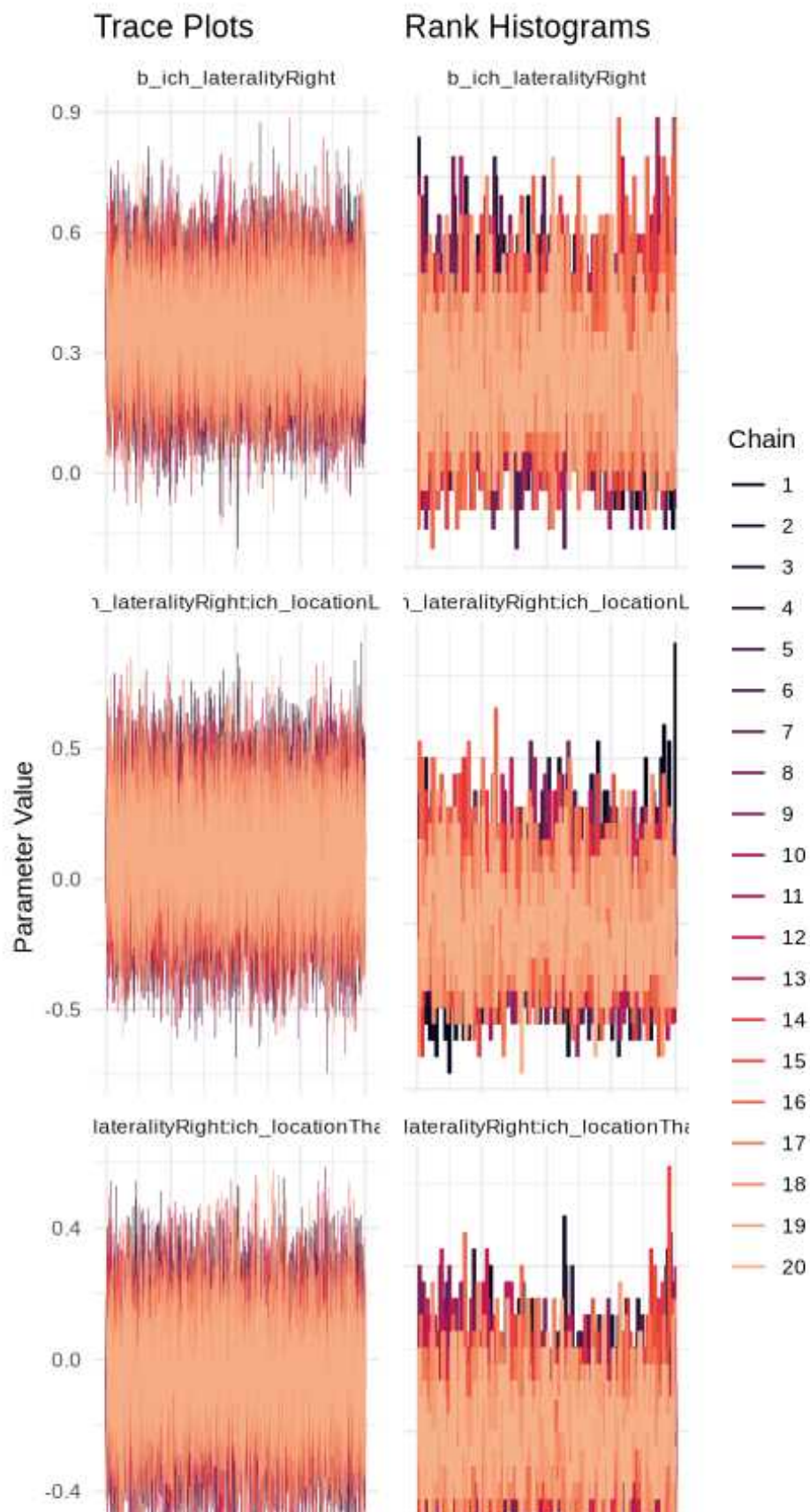
Trace Plots



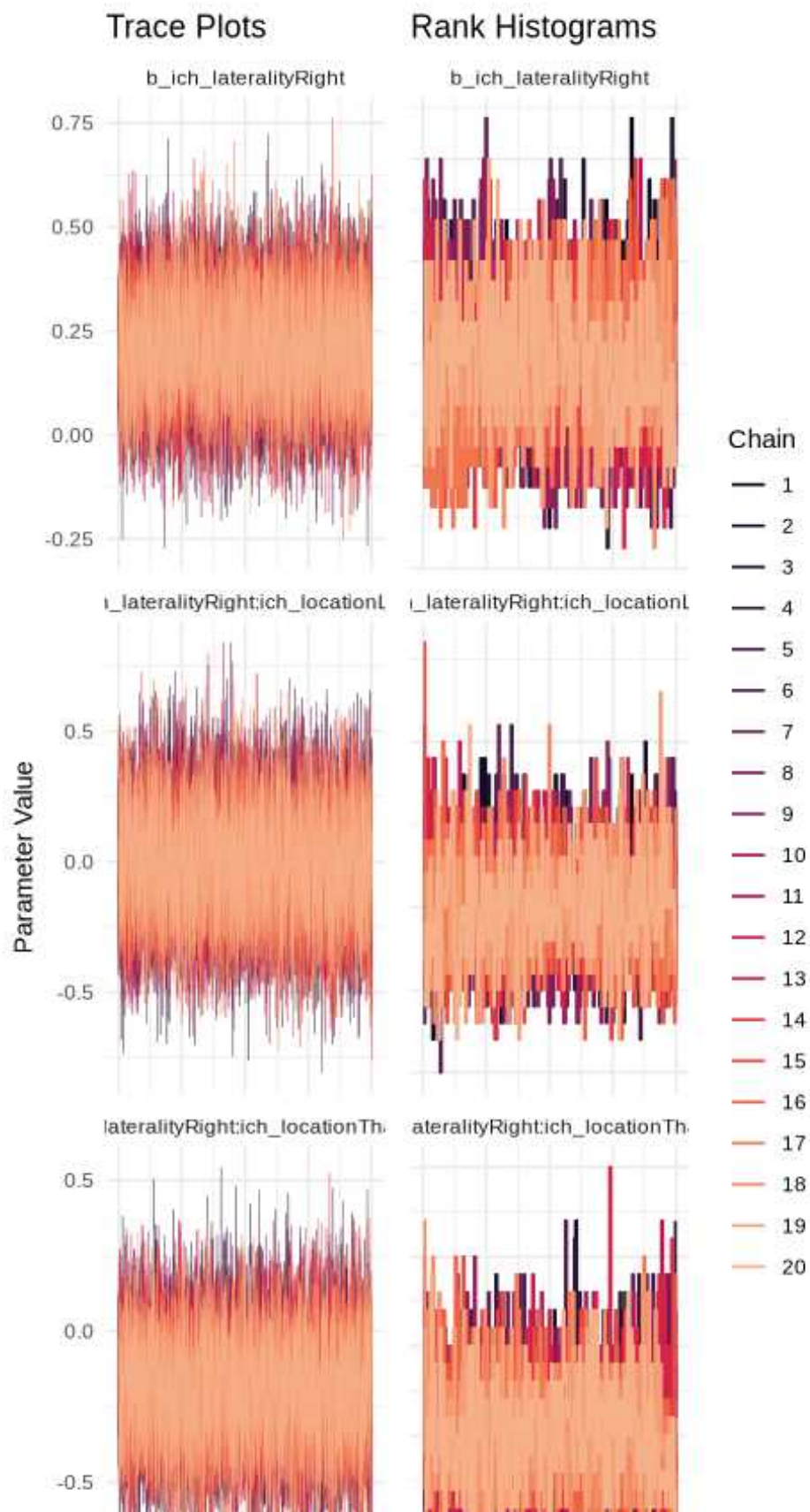
Rank Histograms



Posterior diagnostic checks for EuroQOL — Mobility at 90 Days

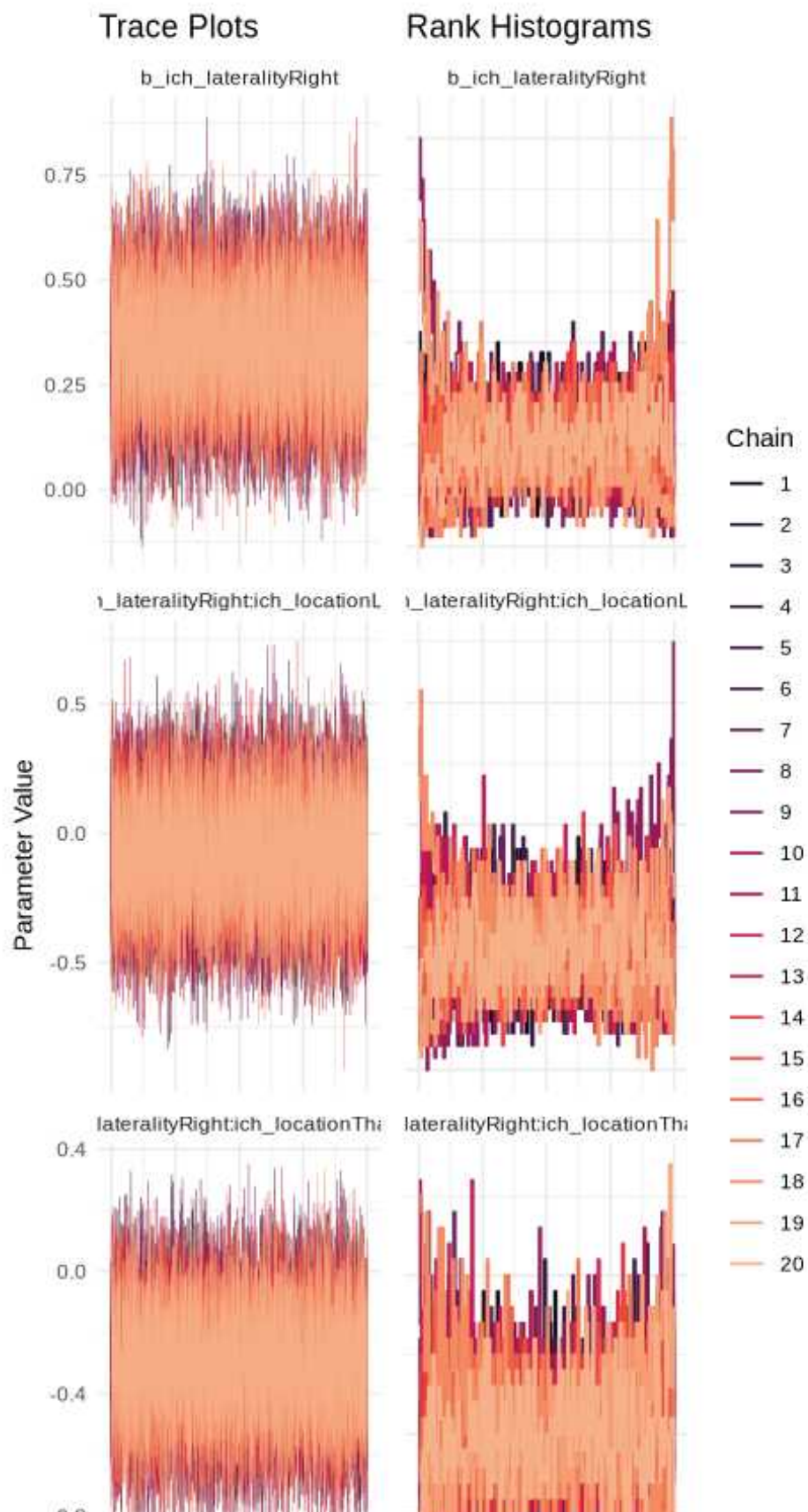


Posterior diagnostic checks for EuroQOL — Self-Care at 90 Days



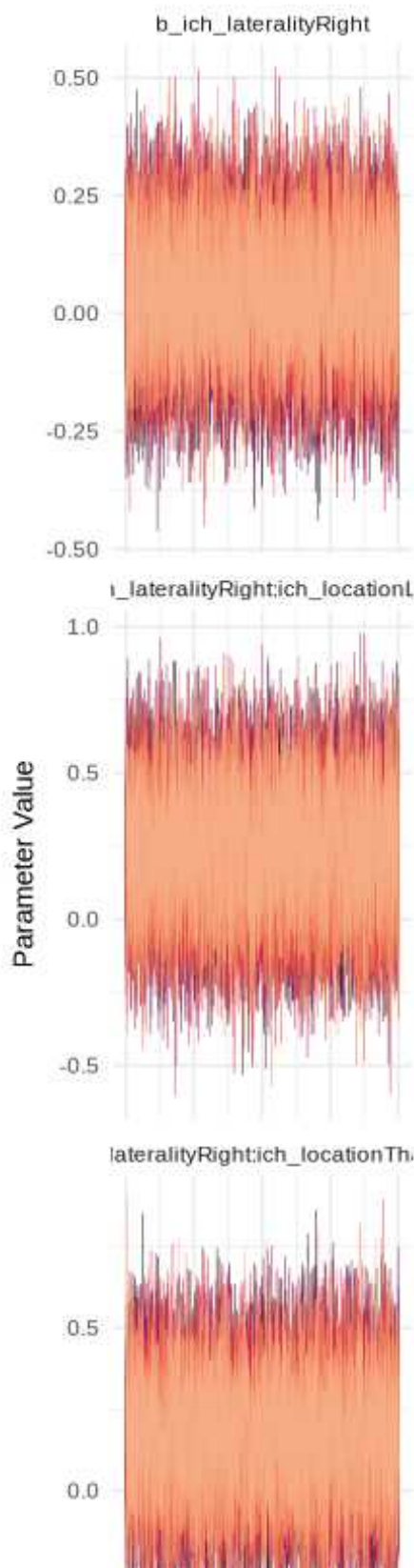
Posterior diagnostic checks for EuroQOL — Usual Activities at 90 Days

EuroQOL Pain/Discomfort 90

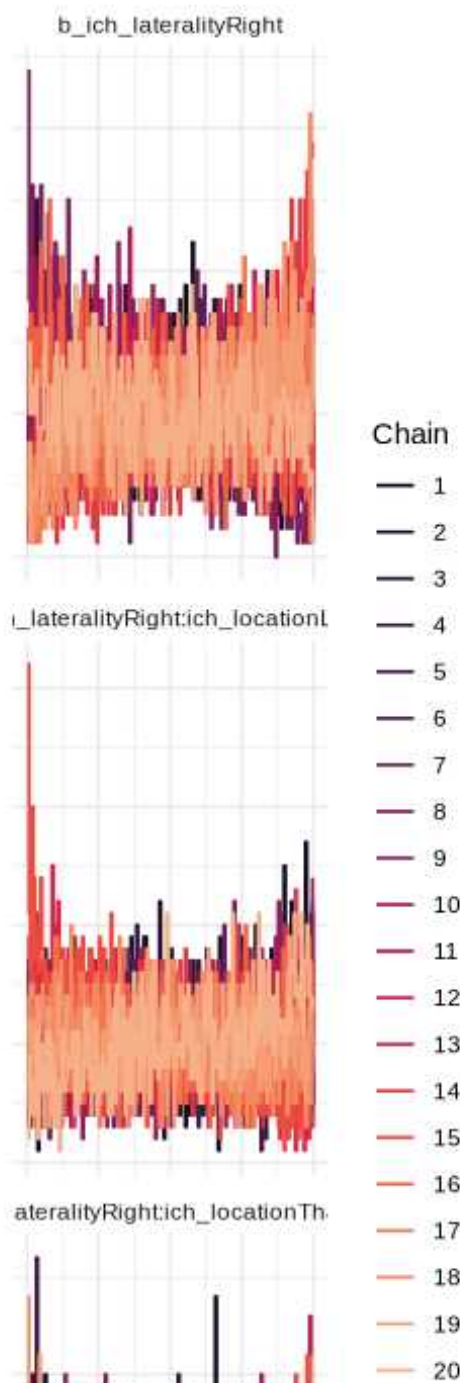


Posterior diagnostic checks for EuroQOL — Pain/Discomfort at 90 Days

Trace Plots

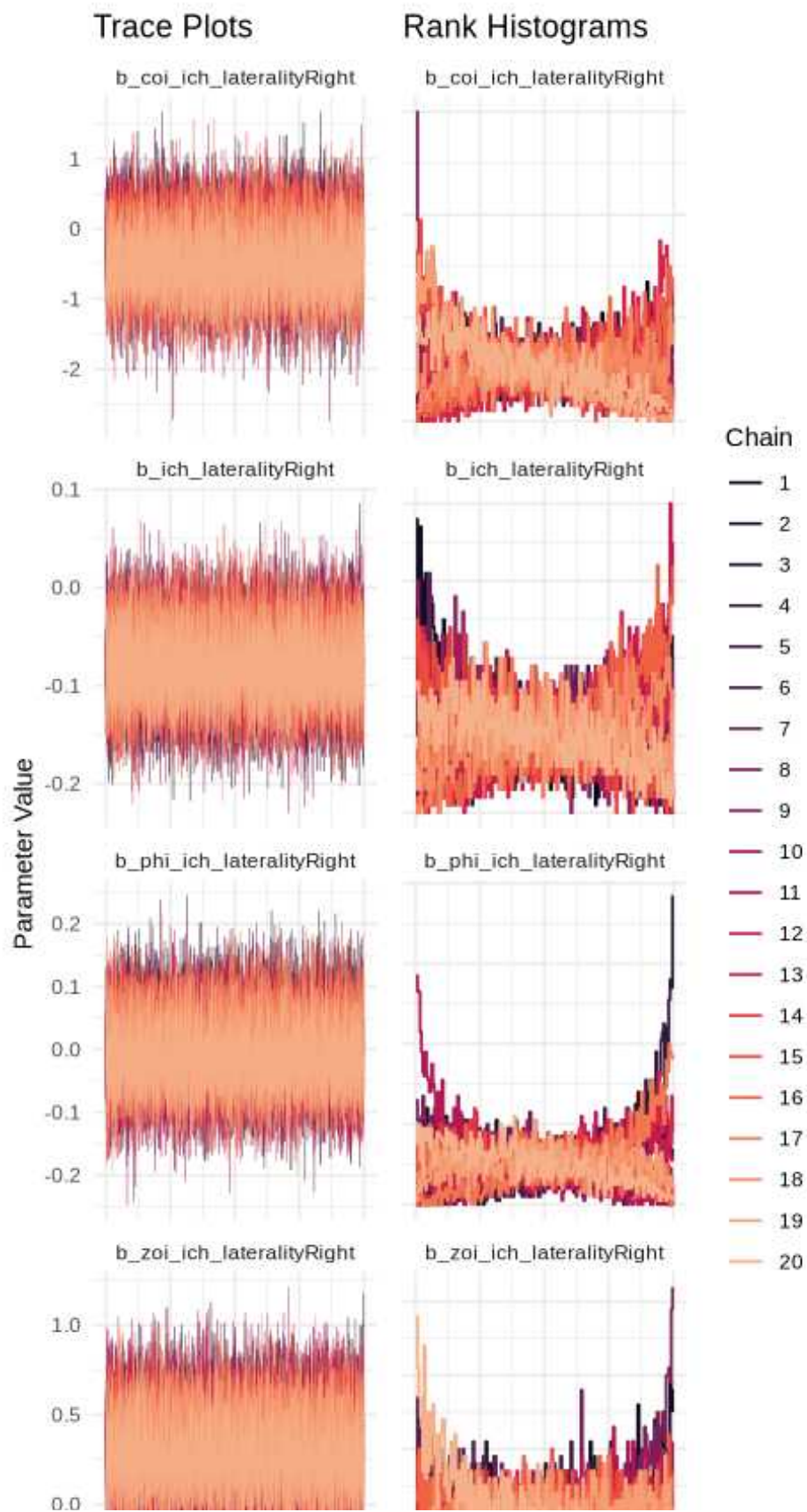


Rank Histograms



Posterior diagnostic checks for EuroQOL — Anxiety/Depression at 90 Days

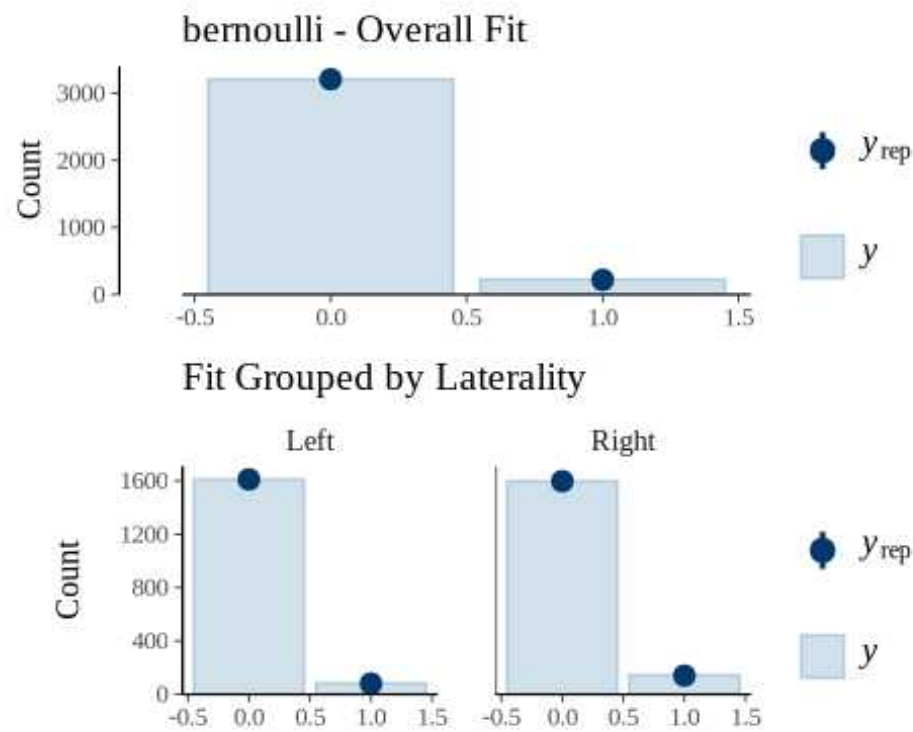
EuroQOL VAS 90



Posterior diagnostic checks for EuroQOL — VAS at 90 Days

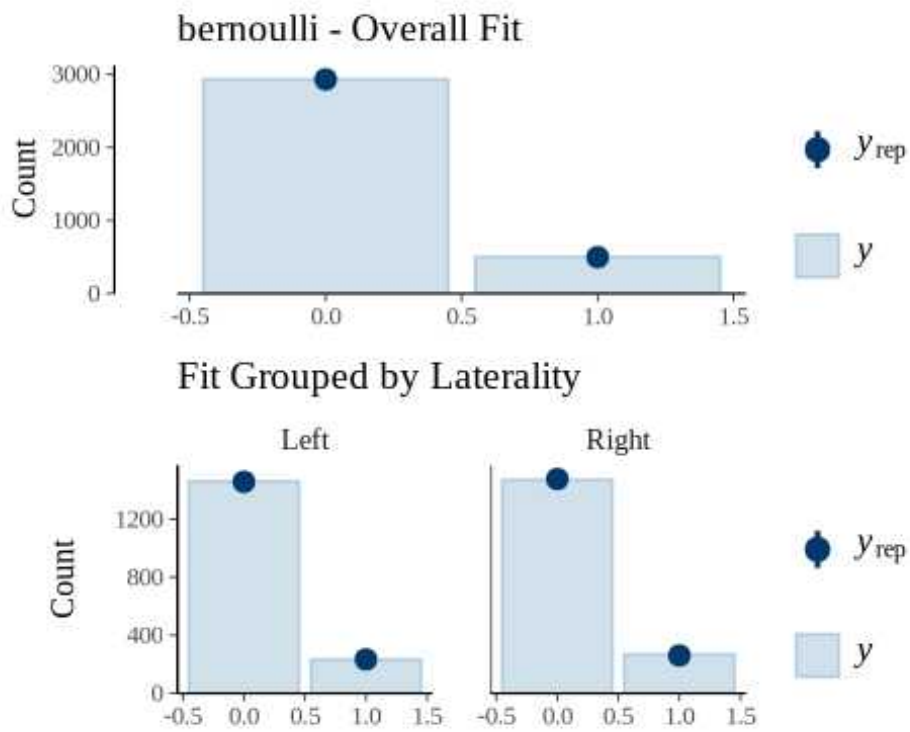
Posterior predictive checks

Neurosurgical Intervention



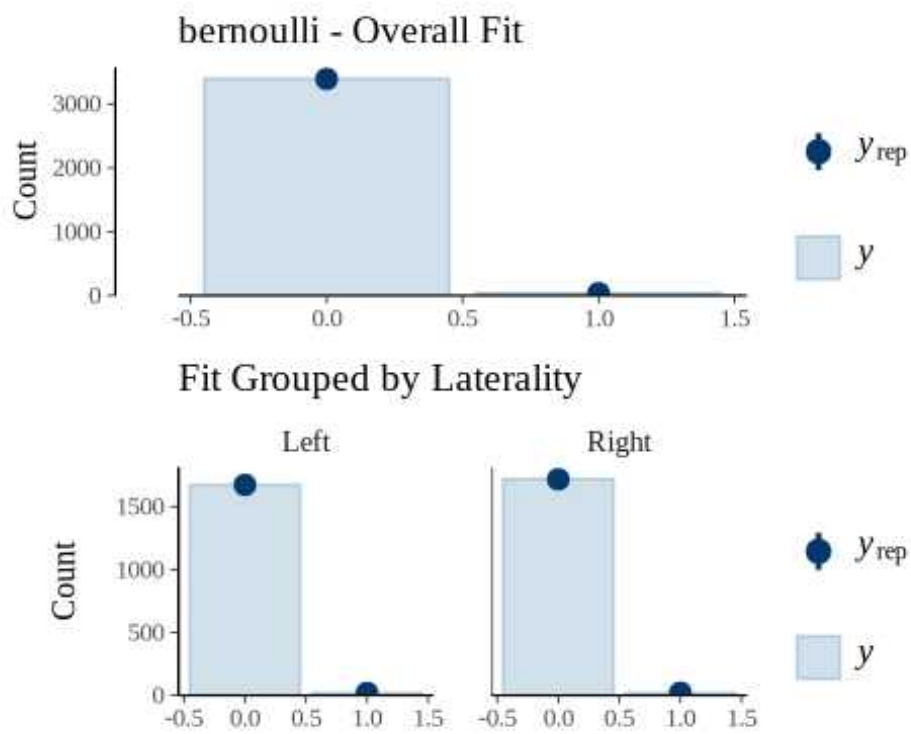
Posterior predictive checks for Neurosurgical Intervention

Ventriculostomy



Posterior predictive checks for Ventriculostomy

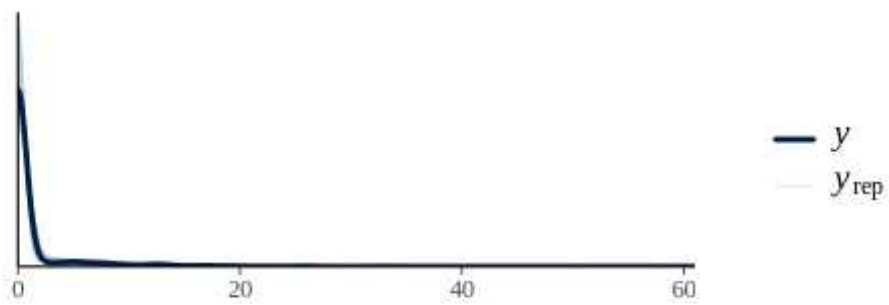
Tracheostomy



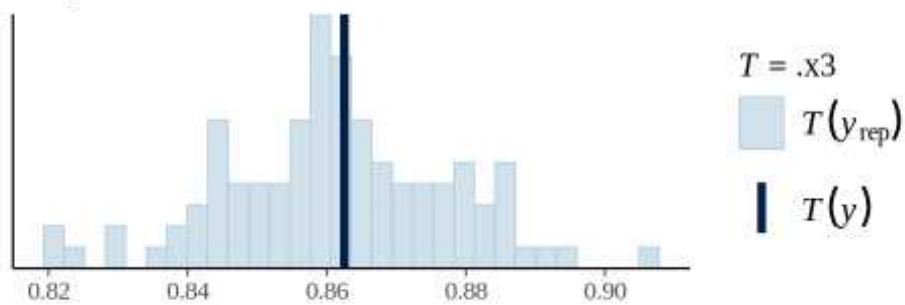
Posterior predictive checks for Tracheostomy

Days of Mechanical Ventilation

zero_inflated_negbinomial - Density Overlay

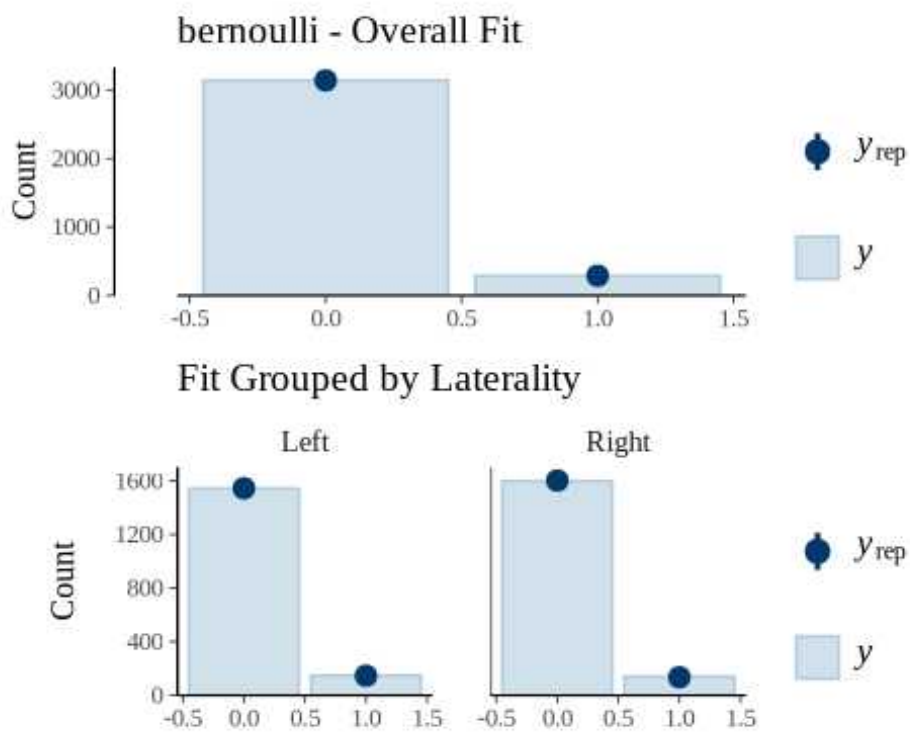


Proportion of Zeroes



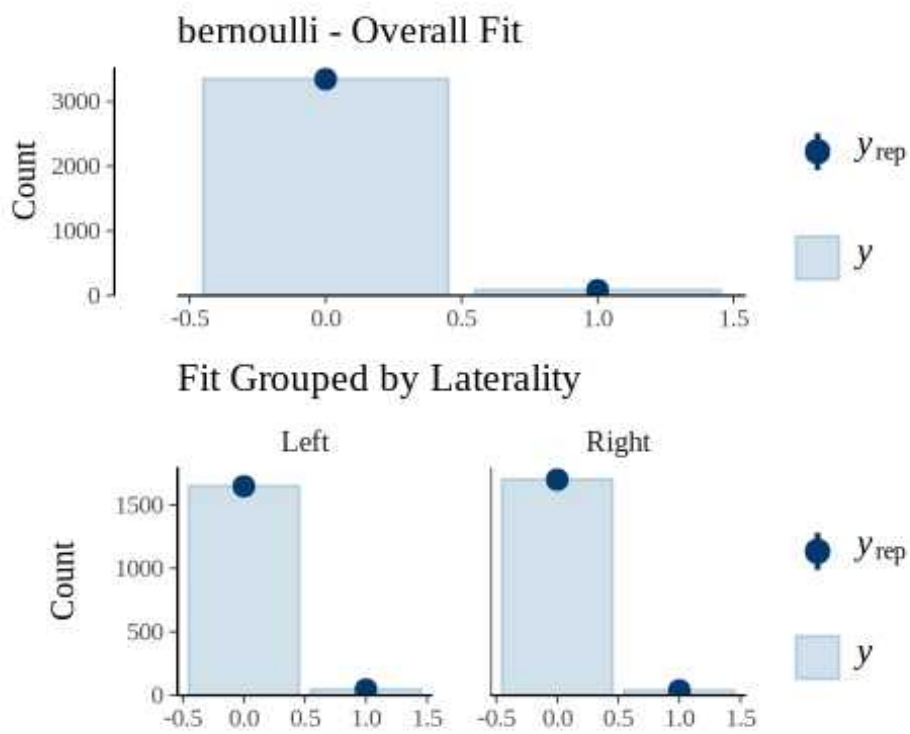
Posterior predictive checks for Days of Mechanical Ventilation

WLST



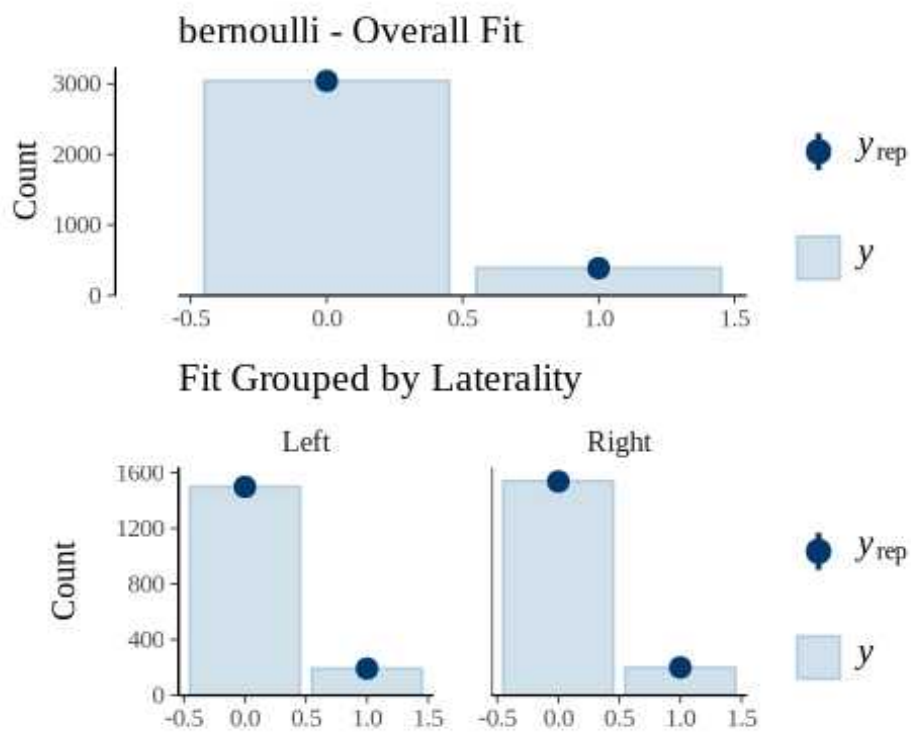
Posterior predictive checks for WLST

Early WLST



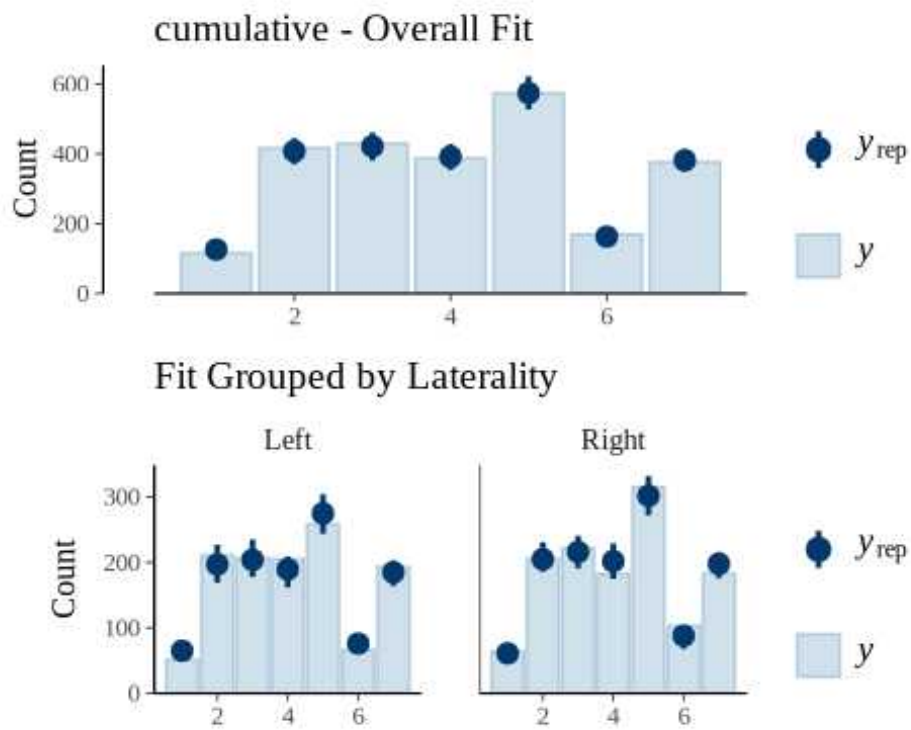
Posterior predictive checks for Early WLST

DNR



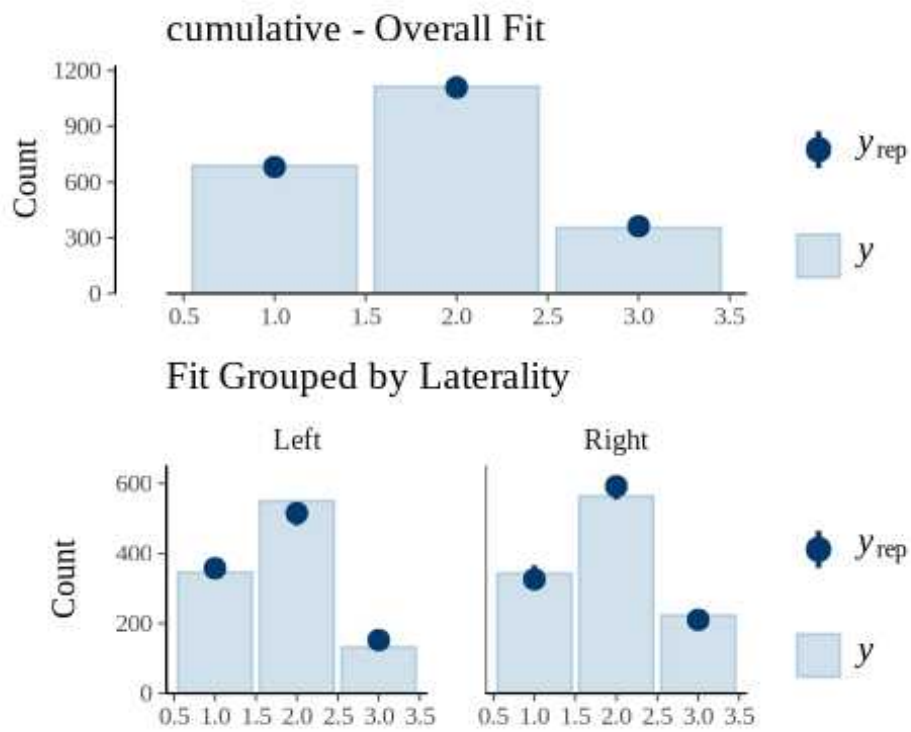
Posterior predictive checks for DNR

mRS 90



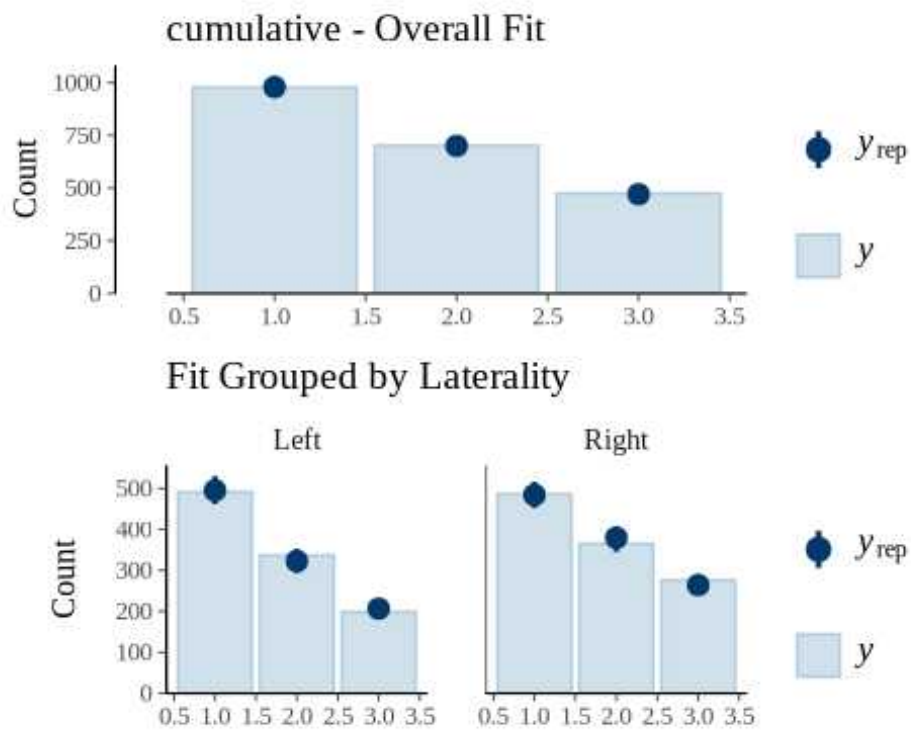
Posterior predictive checks for Modified Rankin Scale at 90 Days

EuroQOL Mobility 90



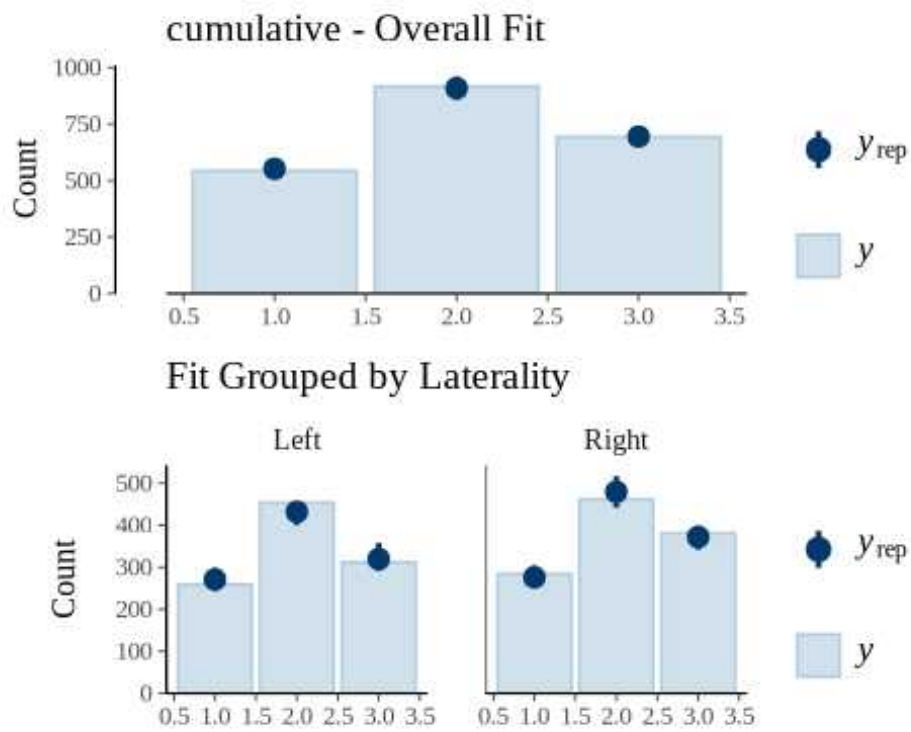
Posterior predictive checks for EuroQOL — Mobility at 90 Days

EuroQOL Self-Care 90



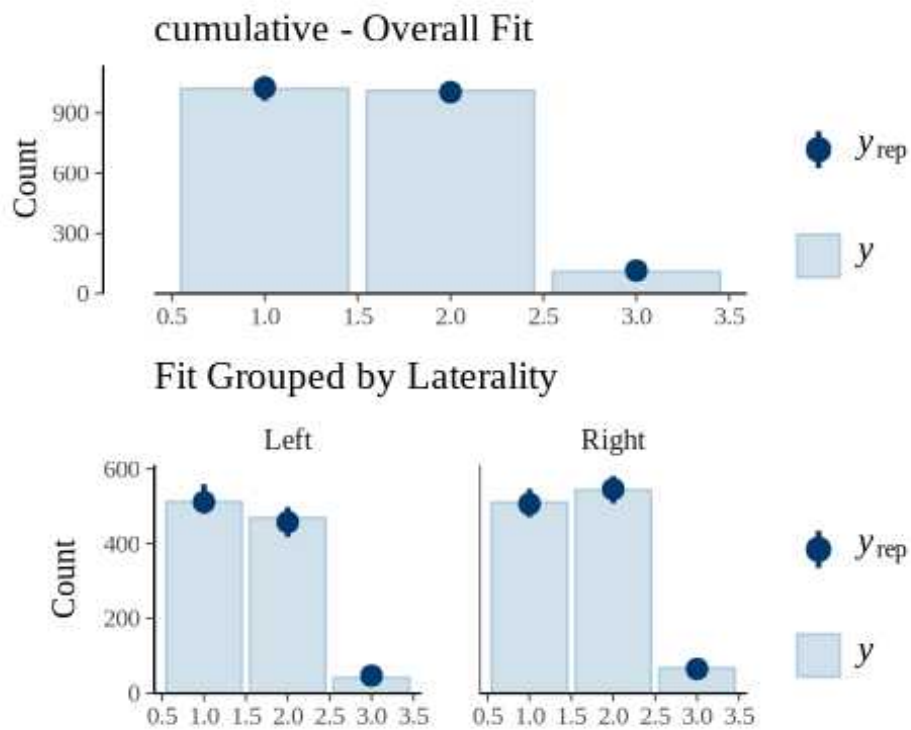
Posterior predictive checks for EuroQOL — Self-Care at 90 Days

EuroQOL Usual Activities 90



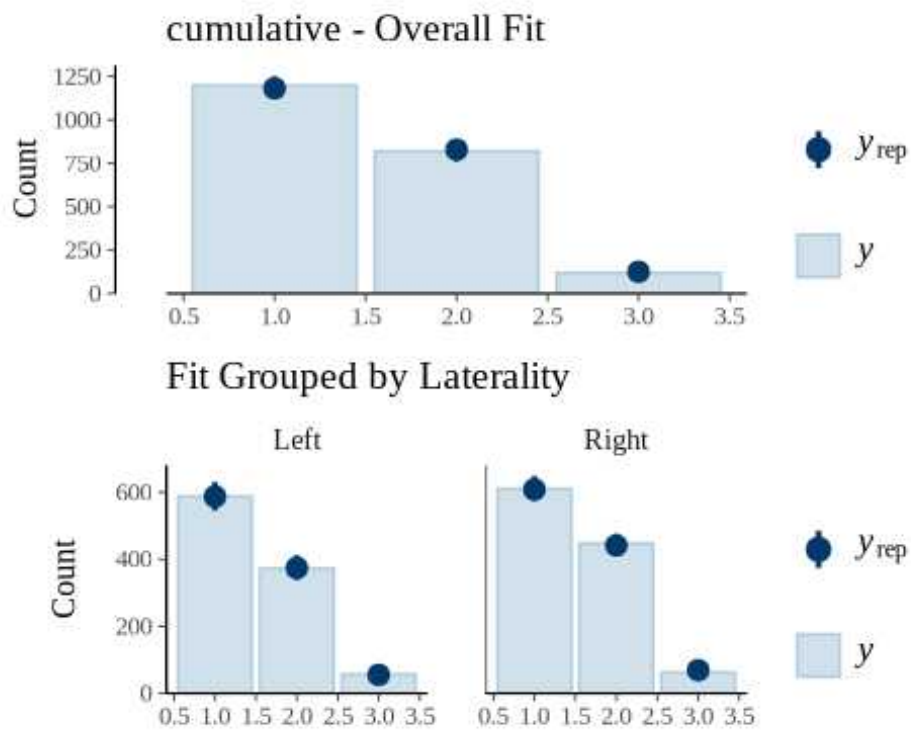
Posterior predictive checks for EuroQOL — Usual Activities at 90 Days

EuroQOL Pain/Discomfort 90

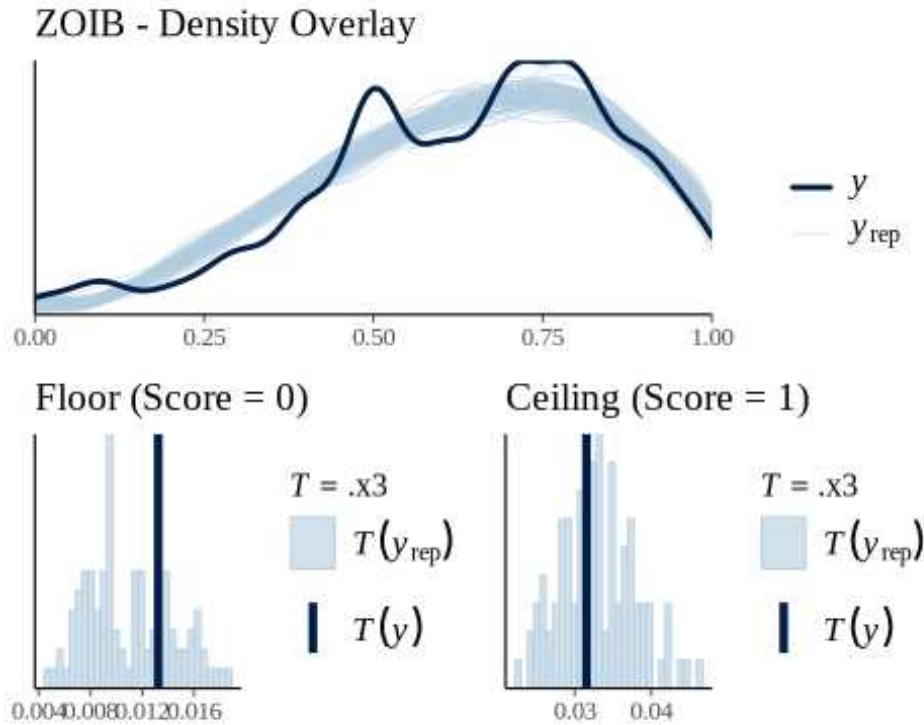


Posterior predictive checks for EuroQOL — Pain/Discomfort at 90 Days

EuroQOL Anxiety/Depression 90



Posterior predictive checks for EuroQOL — Anxiety/Depression at 90 Days



Posterior predictive checks for EuroQOL — VAS at 90 Days

Sensitivity Analyses

To assess the robustness of our findings for the primary outcome, we conducted several sensitivity analyses. First, we repeated the primary analysis using three alternative prior distributions: a left-hemisphere prior, a right-hemisphere prior, and a diffuse (minimally informative) prior. This allowed us to evaluate the influence of different initial beliefs on the resulting posterior distributions. Second, we explored an alternative covariate adjustment set based on a minimal adjustment set identified from our directed acyclic graphs (DAGs) to ensure our results were not dependent on the specific model specification. Third, as described in the primary manuscript, the primary analysis used MICE-imputed data; we performed a complete-case sensitivity analysis (excluding patients with missing outcomes) to confirm that our findings were not unduly influenced by the imputation assumptions. Finally, to evaluate the potential

impact of center-level clustering, we performed a site-level random-effect sensitivity analysis restricted to the ATACH-2 cohort, where hospital site identifiers were available. The results of these sensitivity analyses were directionally consistent with the primary findings.

Exploratory Subgroup Interaction Analysis

To evaluate whether the association between hemispheric laterality and neurosurgical intervention varied across clinically relevant subgroups, we fitted Bayesian logistic regression models that included formal interaction terms between ICH laterality and each subgroup variable. Two interaction models were fitted: (1) laterality $\times$ ICH location (basal ganglia, lobar, thalamic) and (2) laterality $\times$ study (ATACH-2, ERICH). Both models used the neutral prior under the adjusted covariate set and were fitted to complete-case data.

For each subgroup, we report: the subgroup-specific adjusted odds ratio (aOR) for the laterality association with 95% CrI, posterior probabilities of the aOR exceeding 1 and 1.2, the proportion of the posterior within the ROPE, and — critically — the formal interaction effect expressed as a Ratio of Odds Ratios (ROR). The ROR quantifies how many times larger (or smaller) the laterality effect is in a given subgroup compared to the reference subgroup. An ROR of 1 indicates no difference in the laterality association across subgroups, while values substantially above or below 1 suggest effect modification.

This approach avoids the well-known pitfalls of comparing isolated subgroup point estimates, which can yield misleading conclusions about heterogeneity. Instead, the interaction ROR directly tests whether the laterality effect differs between subgroups and provides a formal probabilistic assessment of the degree of heterogeneity.

Table 10: Subgroup Analysis of Hemispheric Laterality and Neurosurgical Intervention with Formal Interaction Tests

Subgroup	Left Hemispheric	Right Hemispheric	aOR (95% CrI) ¹	Posterior Probability of difference (aOR > 1)	Posterior Probability of a substantial difference (aOR > 1.2)	Percentage of posterior within ROPE ²	Interaction ROR (95% CrI) ³	Probability subgroup aOR exceeds reference aOR
ICH Location								
Basal Ganglia	43 (6.2%)	51 (8.3%)	1.54 (1.20 - 1.90)	> 99.9%	96%	< 0.1%	Reference	—
Lobar	39 (7.7%)	76 (14.9%)	1.73 (1.34 - 2.38)	> 99.9%	> 99.9%	< 0.1%	1.13 (0.85 - 1.65)	70%
Thalamus	4 (0.8%)	17 (2.7%)	1.66 (1.06 - 2.48)	98%	92%	4%	1.08 (0.64 - 1.56)	62%
Study								
ATACH-2	13 (2.8%)	30 (5.8%)	1.79 (1.24 - 2.84)	> 99.9%	98%	< 0.1%	Reference	—
ERICH	73 (5.8%)	114 (9.2%)	1.59 (1.25 - 2.14)	> 99.9%	> 99.9%	< 0.1%	0.82 (0.50 - 1.39)	36%

¹aOR = adjusted odds ratio, CrI = 95% credible interval. Reference Category: Left Hemisphere Laterality

²ROPE = region of practical equivalence, defined as 0.95 to 1.05

³Ratio of Odds Ratios (ROR). Represents how many times larger the laterality effect (aOR) is in this subgroup compared to the Reference subgroup.

The laterality association with neurosurgical intervention was directionally consistent across ICH location and study subgroups. While the point estimates suggested a potentially larger laterality effect for lobar hemorrhages relative to basal ganglia hemorrhages, the interaction ROR credible intervals were wide and included 1, indicating that the observed subgroup differences were not

statistically distinguishable. Similarly, the laterality effect was consistent between the ATACH-2 and ERICH cohorts, with no evidence of meaningful effect modification by study source.

Table 2 with Different Priors

Left Hemisphere Prior

Table 11

Outcome	Left Hemisphere	Right Hemisphere	aOR / IRR (95% CrI)	Posterior Probability of difference (Est > 1)	Posterior Probability of substantial difference (Est > 1.2)	Percentage of posterior within ROPE
Neurosurgical Intervention	86 (5.0%)	144 (8.2%)	1.38 (1.12 - 1.71)	99.9%	90.2%	0.6%
External Ventricular Drain	236 (13.7%)	272 (15.5%)	1.13 (0.97 - 1.3)	94.1%	19.8%	16.7%
Days of Mechanical Ventilation	6 (3 – 12)	6 (3 – 9)	0.82 (0.64 - 1.04)	5.4%	< 0.1%	10.1%
Tracheostomy	19 (1.1%)	19 (1.1%)	0.96 (0.67 - 1.37)	40.3%	11%	20.7%
Withdrawal of Life-Sustaining Therapy	152 (8.8%)	143 (8.1%)	0.9 (0.74 - 1.08)	12.8%	0.1%	22.4%
Early Withdrawal of Life-Sustaining Therapy	45 (2.6%)	39 (2.2%)	0.89 (0.65 - 1.22)	24.2%	3.3%	19.6%
DNR Order	195 (11.3%)	202 (11.5%)	0.96 (0.82 - 1.13)	32.4%	0.4%	42%

Values represent Average Marginal Effects (Odds Ratios for binary outcomes, Incidence Rate Ratios for counts). Models are adjusted for age, admission GCS, ICH location (interaction), ICH volume, IVH, and study. Days of mechanical ventilation represent median (IQR) days (data from ATACH-2 only). ROPE indicates region of practical equivalence (0.95 to 1.05); aOR, adjusted odds ratio; IRR, incidence rate ratio; CrI, credible interval; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; IQR, interquartile range; and IVH, intraventricular hemorrhage.

Right Hemisphere Prior

Table 12

Outcome	Left Hemisphere	Right Hemisphere	aOR / IRR (95% CrI)	Posterior Probability of difference (Est > 1)	Posterior Probability of substantial difference (Est > 1.2)	Percentage of posterior within ROPE
Neurosurgical Intervention	86 (5.0%)	144 (8.2%)	1.55 (1.26 - 1.93)	> 99.9%	99.1%	< 0.1%
External Ventricular Drain	236 (13.7%)	272 (15.5%)	1.2 (1.04 - 1.39)	99.2%	51.1%	3.7%
Days of Mechanical Ventilation	6 (3 – 12)	6 (3 – 9)	1.01 (0.79 - 1.32)	53.7%	10.3%	30.2%
Tracheostomy	19 (1.1%)	19 (1.1%)	1.22 (0.85 - 1.75)	86.3%	53.8%	11.8%
Withdrawal of Life-Sustaining Therapy	152 (8.8%)	143 (8.1%)	0.98 (0.81 - 1.18)	39.8%	1.6%	38.3%
Early Withdrawal of Life-Sustaining Therapy	45 (2.6%)	39 (2.2%)	1.04 (0.76 - 1.41)	59.1%	17.6%	24.7%
DNR Order	195 (11.3%)	202 (11.5%)	1.03 (0.88 - 1.21)	65%	3.4%	42.8%

Values represent Average Marginal Effects (Odds Ratios for binary outcomes, Incidence Rate Ratios for counts). Models are adjusted for age, admission GCS, ICH location (interaction), ICH volume, IVH, and study. Days of mechanical ventilation represent median (IQR) days (data from ATACH-2 only). ROPE indicates region of practical equivalence (0.95 to 1.05); aOR, adjusted odds ratio; IRR, incidence rate ratio; CrI, credible interval; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; IQR, interquartile range; and IVH, intraventricular hemorrhage.

Diffuse Prior

Table 13

Outcome	Left Hemisphere	Right Hemisphere	aOR / IRR (95% CrI)	Posterior Probability of difference (Est > 1)	Posterior Probability of substantial difference (Est > 1.2)	Percentage of posterior within ROPE
Neurosurgical Intervention	86 (5.0%)	144 (8.2%)	1.73 (1.33 - 2.26)	> 99.9%	99.6%	< 0.1%
External Ventricular Drain	236 (13.7%)	272 (15.5%)	1.23 (1.04 - 1.46)	99.2%	62.6%	2.9%
Days of Mechanical Ventilation	6 (3 – 12)	6 (3 – 9)	0.94 (0.71 - 1.27)	34.1%	5.5%	24.6%
Tracheostomy	19 (1.1%)	19 (1.1%)	1.17 (0.61 - 2.21)	67.9%	46.6%	10.9%
Withdrawal of Life-Sustaining Therapy	152 (8.8%)	143 (8.1%)	0.95 (0.76 - 1.18)	30.7%	1.7%	31.2%
Early Withdrawal of Life-Sustaining Therapy	45 (2.6%)	39 (2.2%)	0.9 (0.59 - 1.36)	30.4%	8.5%	16.5%
DNR Order	195 (11.3%)	202 (11.5%)	1.01 (0.84 - 1.22)	55.8%	3.7%	40.4%

Values represent Average Marginal Effects (Odds Ratios for binary outcomes, Incidence Rate Ratios for counts). Models are adjusted for age, admission GCS, ICH location (interaction), ICH volume, IVH, and study. Days of mechanical ventilation represent median (IQR) days (data from ATACH-2 only). ROPE indicates region of practical equivalence (0.95 to 1.05); aOR, adjusted odds ratio; IRR, incidence rate ratio; CrI, credible interval; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; IQR, interquartile range; and IVH, intraventricular hemorrhage.

Table 3 with Different Priors

Left Hemisphere Prior

Table 14

Outcome	aOR / Mean Diff (95% CrI)	Posterior Probability of any difference	Posterior Probability of a substantial difference	Percentage of posterior within ROPE
Modified Rankin Score	0.99 (0.98 - 1.00)	3.2%	< 0.1%	> 99.9%
EuroQOL - Mobility	1.03 (1.01 - 1.04)	> 99.9%	< 0.1%	99.8%
EuroQOL - Self-Care	1.02 (1.01 - 1.04)	> 99.9%	< 0.1%	99.8%
EuroQOL - Usual Activities	1.00 (0.99 - 1.00)	13.1%	< 0.1%	> 99.9%
EuroQOL - Pain/Discomfort	1.03 (0.99 - 1.08)	94.1%	< 0.1%	75.4%
EuroQOL - Anxiety/Depression	1.03 (0.99 - 1.08)	90.2%	< 0.1%	81.8%
EQ-VAS	-1.8 (-3.7 - 0.0)	97.3%	< 0.1%	56.6%

Values represent Average Marginal Effects (Odds Ratios for ordinal outcomes; Mean Difference in points [0–100] for EQ-VAS). Models are adjusted for age, admission GCS, ICH location, ICH volume, IVH, and study (as random intercept); Reference Category: Left Hemisphere. ROPE indicates region of practical equivalence (0.95 to 1.05 for ordinal; ± 2 points for EQ-VAS). For EQ-VAS, probability of difference is defined as Mean Difference < 0, and substantial difference as < -5 points. aOR indicates adjusted odds ratio; CrI, credible interval; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; and IVH, intraventricular hemorrhage.

Right Hemisphere Prior

Table 15

Outcome	aOR / Mean Diff (95% CrI)	Posterior Probability of any difference	Posterior Probability of a substantial difference	Percentage of posterior within ROPE
Modified Rankin Score	0.99 (0.98 - 1.00)	0.3%	< 0.1%	> 99.9%
EuroQOL - Mobility	1.03 (1.01 - 1.05)	> 99.9%	< 0.1%	98.7%
EuroQOL - Self-Care	1.03 (1.02 - 1.05)	> 99.9%	< 0.1%	98.6%
EuroQOL - Usual Activities	0.99 (0.99 - 1.00)	2.2%	< 0.1%	> 99.9%
EuroQOL - Pain/Discomfort	1.05 (1.01 - 1.10)	98.9%	< 0.1%	46.4%
EuroQOL - Anxiety/Depression	1.05 (1.00 - 1.09)	98.2%	< 0.1%	54.1%
EQ-VAS	-1.5 (-3.3 - 0.4)	94.2%	< 0.1%	70.9%

Values represent Average Marginal Effects (Odds Ratios for ordinal outcomes; Mean Difference in points [0–100] for EQ-VAS). Models are adjusted for age, admission GCS, ICH location, ICH volume, IVH, and study (as random intercept); Reference Category: Left Hemisphere. ROPE indicates region of practical equivalence (0.95 to 1.05 for ordinal; ± 2 points for EQ-VAS). For EQ-VAS, probability of difference is defined as Mean Difference < 0, and substantial difference as < -5 points. aOR indicates adjusted odds ratio; CrI, credible interval; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; and IVH, intraventricular hemorrhage.

Diffuse Prior

Table 16

Outcome	aOR / Mean Diff (95% CrI)	Posterior Probability of any difference	Posterior Probability of a substantial difference	Percentage of posterior within ROPE
Modified Rankin Score	0.99 (0.98 - 1.00)	0.5%	< 0.1%	> 99.9%
EuroQOL - Mobility	1.03 (1.01 - 1.05)	> 99.9%	< 0.1%	98.2%
EuroQOL - Self-Care	1.03 (1.02 - 1.05)	> 99.9%	< 0.1%	95.6%
EuroQOL - Usual Activities	0.99 (0.99 - 1.00)	2.4%	< 0.1%	> 99.9%
EuroQOL - Pain/Discomfort	1.06 (1.01 - 1.11)	99.2%	< 0.1%	35.9%
EuroQOL - Anxiety/Depression	1.04 (0.99 - 1.09)	95.3%	< 0.1%	63.4%
EQ-VAS	-1.7 (-3.6 - 0.2)	96.1%	< 0.1%	62.5%

Values represent Average Marginal Effects (Odds Ratios for ordinal outcomes; Mean Difference in points [0–100] for EQ-VAS). Models are adjusted for age, admission GCS, ICH location, ICH volume, IVH, and study (as random intercept); Reference Category: Left Hemisphere. ROPE indicates region of practical equivalence (0.95 to 1.05 for ordinal; ± 2 points for EQ-VAS). For EQ-VAS, probability of difference is defined as Mean Difference < 0, and substantial difference as < -5 points. aOR indicates adjusted odds ratio; CrI, credible interval; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; and IVH, intraventricular hemorrhage.

Complete-Case Sensitivity Analysis

The primary analysis used multiple imputation by chained equations (MICE) with 20 imputed datasets, as described in the main manuscript. As a sensitivity analysis, we repeated the primary models using a complete-case approach (excluding patients with missing outcomes) under the neutral prior with the adjusted covariate set. These models were fitted to the complete-case data for all primary, secondary, and functional outcomes. All models were fitted using Markov Chain Monte Carlo (MCMC) with 4 chains and 5,000 iterations per chain (including a warmup period of 1,000 iterations).

Table 17: Outcomes of Interest Using Complete-Case Analysis (Sensitivity)

Outcome	Left Hemisphere	Right Hemisphere	aOR / IRR (95% CrI)	Posterior Probability of difference (Est > 1)	Posterior Probability of substantial difference (Est > 1.2)	Percentage of posterior within ROPE
Neurosurgical Intervention	86 (5.0%)	144 (8.2%)	1.65 (1.28 - 2.13)	> 99.9%	99.3%	< 0.1%
External Ventricular Drain	236 (13.7%)	272 (15.5%)	1.22 (1.04 - 1.44)	99%	58.7%	3.3%
Days of Mechanical Ventilation	6 (3 – 12)	6 (3 – 9)	0.99 (0.89 - 1.09)	39%	< 0.1%	65.6%
Tracheostomy	19 (1.1%)	19 (1.1%)	1.12 (0.66 - 1.94)	66.8%	40.7%	13.5%
Withdrawal of Life-Sustaining Therapy	152 (8.8%)	143 (8.1%)	0.95 (0.76 - 1.17)	30.2%	1.5%	31.4%
Early Withdrawal of Life-Sustaining Therapy	45 (2.6%)	39 (2.2%)	0.92 (0.64 - 1.34)	33.3%	8.6%	19.3%
DNR Order	195 (11.3%)	202 (11.5%)	1.01 (0.84 - 1.21)	54.9%	3%	42.2%

Values represent Average Marginal Effects (Odds Ratios for binary outcomes, Incidence Rate Ratios for counts). Models are adjusted for age, admission GCS, ICH location (interaction), ICH volume, IVH, and study. Days of mechanical ventilation represent median (IQR) days (data from ATACH-2 only). ROPE indicates region of practical equivalence (0.95 to 1.05); aOR, adjusted odds ratio; IRR, incidence rate ratio; CrI, credible interval; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; IQR, interquartile range; and IVH, intraventricular hemorrhage.

Table 18: Functional Outcomes of Interest Using Complete-Case Analysis (Sensitivity)

Outcome	aOR / Mean Diff (95% CrI)	Posterior Probability of any difference	Posterior Probability of a substantial difference	Percentage of posterior within ROPE
Modified Rankin Score	0.99 (0.98 - 1.00)	0.7%	< 0.1%	> 99.9%
EuroQOL - Mobility	1.04 (1.02 - 1.07)	> 99.9%	< 0.1%	75.6%
EuroQOL - Self-Care	1.04 (1.02 - 1.06)	> 99.9%	< 0.1%	78.1%
EuroQOL - Usual Activities	1.00 (0.99 - 1.00)	2.6%	< 0.1%	> 99.9%
EuroQOL - Pain/Discomfort	1.06 (1.01 - 1.11)	98.9%	< 0.1%	40.9%
EuroQOL - Anxiety/Depression	1.04 (0.99 - 1.09)	95.4%	< 0.1%	63.4%
EQ-VAS	-1.5 (-3.4 - 0.3)	95.1%	< 0.1%	69.2%

Values represent Average Marginal Effects (Odds Ratios for ordinal outcomes; Mean Difference in points [0–100] for EQ-VAS). Models are adjusted for age, admission GCS, ICH location, ICH volume, IVH, and study (as random intercept); Reference Category: Left Hemisphere. ROPE indicates region of practical equivalence (0.95 to 1.05 for ordinal; ± 2 points for EQ-VAS). For EQ-VAS, probability of difference is defined as Mean Difference < 0, and substantial difference as < -5 points. aOR indicates adjusted odds ratio; CrI, credible interval; GCS, Glasgow Coma Scale; ICH, intracerebral hemorrhage; and IVH, intraventricular hemorrhage.

Site-Level Clustering Sensitivity Analysis

Neurosurgical decision-making is highly influenced by local practice patterns, institutional resources, and surgeon availability. Our primary models included a random intercept for study (ATACH-2 versus ERICH) to account for systematic differences between the two cohorts, but this two-level random effect is unlikely to fully capture the heterogeneity in clinical practice across the many individual hospital sites within each study.

Ideally, we would have included a random intercept for hospital site in all models. However, site-level identifiers were only available for the ATACH-2 cohort and were not included in the ERICH dataset. To assess the potential impact of site-level clustering on our primary finding, we

performed a sensitivity analysis restricted to the ATACH-2 cohort, where we fitted two versions of the primary model for neurosurgical intervention:

1. **Base model:** The same model specification as the primary analysis (adjusted for ICH location, age, GCS, ICH volume, IVH, and an ICH laterality-by-location interaction) but without any random intercept, since ATACH-2 is a single study.
2. **Site random-effect model:** The same model specification with the addition of a random intercept for hospital site, allowing the baseline probability of neurosurgical intervention to vary across sites.

Both models were fitted using multiple imputation (20 imputed datasets) with the neutral prior under the adjusted covariate set. The table below compares the posterior estimates for the association between hemispheric laterality and neurosurgical intervention from both models.

Table 19: ATACH-2 Sensitivity Analysis: Association of Hemispheric Laterality with Neurosurgical Intervention With and Without Site-Level Random Intercept

ATACH-2 Model Variants	Left Hemisphere	Right Hemisphere	aOR (95% CrI) ¹	Posterior Probability of difference (aOR > 1)	Posterior Probability of a substantial difference (aOR > 1.2)	Percentage of posterior within ROPE
ATACH-2 (Base Model)	13 (2.8%)	30 (5.8%)	2.23 (1.31 - 3.76)	99.8%	98.6%	< 0.1%
ATACH-2 (Site Random Effect)	13 (2.8%)	30 (5.8%)	1.86 (1.10 - 3.16)	99%	94.7%	1.2%

¹aOR = adjusted odds ratio, CrI = 95% credible interval.

Inclusion of the site-level random intercept modestly attenuated the point estimate for the laterality association, which is expected given the additional variance component and the smaller sample size of the ATACH-2 cohort alone. However, the direction, magnitude, and posterior

probability of the association remained consistent with the primary analysis, suggesting that site-level clustering does not materially alter the principal finding.

References and Software Bibliography

The statistical analysis and data processing for this manuscript were conducted using the R programming language. To facilitate reproducibility and to credit the developers of the open-source tools utilized in this research, a complete software bibliography was automatically generated using the `softbib` package.

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